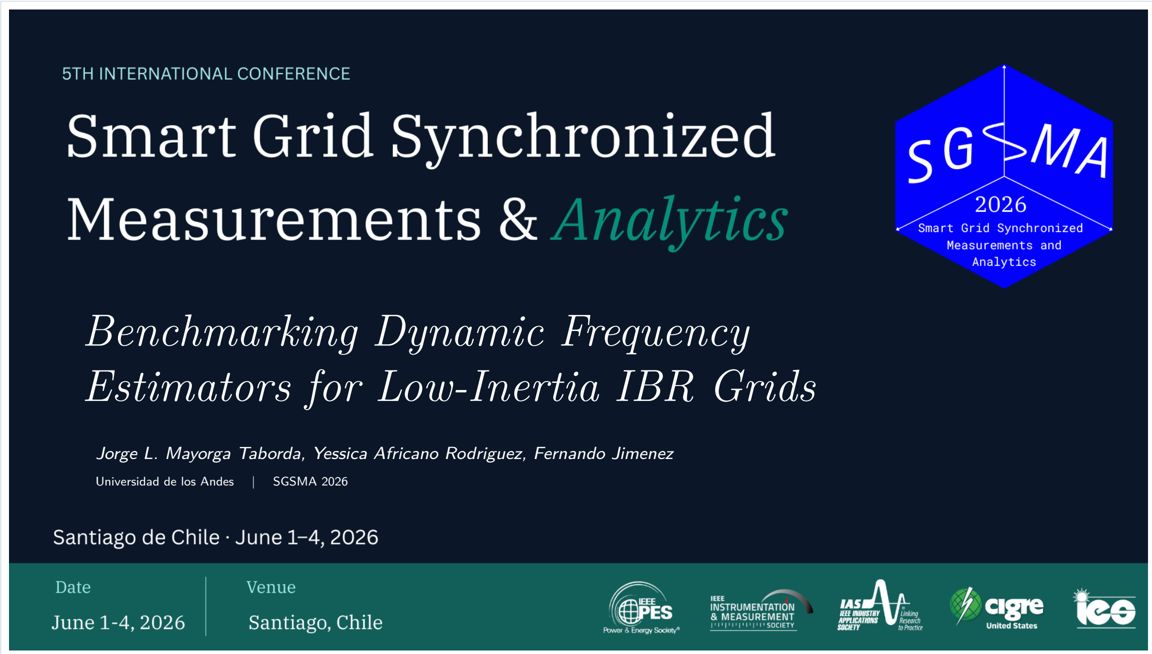

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Opening

Purpose. Set the thesis before showing any data.

Speaker notes

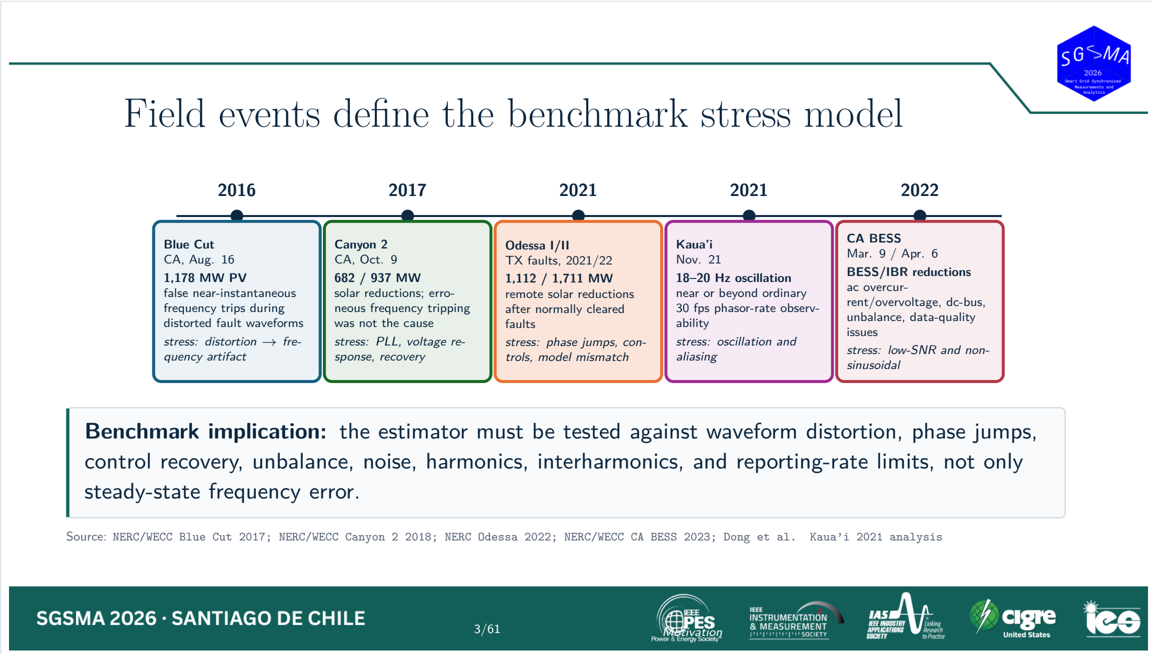
- Good morning. I will discuss dynamic frequency estimation for low-inertia IBR grids.
- The central point is that frequency estimation is no longer a single-accuracy problem. It is a stress-, metric-, and deployment-conditioned problem.
- Preview the answer: no universal estimator wins; the useful output is a qualification map.

Motivation

Purpose. Move from title to problem.

Speaker notes

- Low-inertia grids expose frequency estimators to fast controls, distorted waveforms, and recovery dynamics.
- Say this plainly: the estimator must survive the event, not only look accurate after the event.
- Transition: field events show why this matters.



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Field events

Purpose. Use real events to justify the stress model.

Speaker notes

- These events are not used as claims of prevention. They define stress modes seen in practice.
- Blue Cut and Canyon 2 motivate distorted faults and ride-through behavior; Odessa motivates remote IBR reductions and model mismatch; Kauai motivates observability limits; CA BESS motivates low-SNR and non-sinusoidal data.
- Key line: the benchmark stress set is derived from physical failure modes, not arbitrary waveforms.

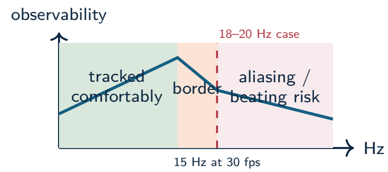
Standard PMU streams show consequences, not always causes

Fast fault and ride-through

IBR controls can react within cycles. A low-rate or heavily filtered frequency estimate may miss the subcycle cause while still showing the delayed consequence.

Oscillation near reporting-rate limits

The Kaua'i Island event on Nov. 21, 2021 exhibited 18–20 Hz oscillations after an oil-power-plant trip. A 30 frame/s phasor stream has a 15 Hz Nyquist limit for phasor-time variations, so oscillations near this range can alias or appear as beating.



Harmonics and interharmonics

Standard synchrophasor outputs summarize the fundamental component; harmonic and interharmonic contamination can still dominate estimator failure modes.

PMU observability limits

Purpose. Explain why ordinary PMU streams may be late.

Speaker notes

- Standard PMU streams are excellent for consequences, but not always for sub-cycle causes.
- The Kauai 18–20 Hz oscillation example is important because 30 fps reporting can alias phasor-time variations near that range.
- Do not overclaim: say this motivates high-rate estimation and stress testing, not that a PMU alone would have prevented an event.

Research problem

- Low-inertia IBR grids require reliable local frequency estimates during ramps, phase jumps, harmonic distortion, ringdown, and mixed dynamic events.
- Classical estimators often trade off responsiveness, noise rejection, latency, and computational load.
- Modern model-based and data-driven estimators improve some regimes, but can be difficult to compare fairly.
- Existing comparisons often under-specify seeds, disturbance families, tuning policy, trip-risk, structural latency, or computational cost.

Research gap

The literature needs a reproducible benchmark that connects estimator error to protection-oriented risk, settling behavior, and deployability.

Research problem

Purpose. State the gap in one sentence.

Speaker notes

- Low-inertia IBR grids require reliable local frequency estimates during ramps, phase jumps, harmonics, and mixed dynamic events.
- Existing comparisons often change seeds, tuning, scenarios, or metrics, so rankings are hard to interpret.
- The gap: a reproducible benchmark that connects estimator error to trip-risk, cost, and deployability.

Operating claim

- Frequency estimation under grid disturbances is a **multi-objective measurement problem**.
- Rankings change with the objective: frequency RMSE, RoCoF error, latency, CPU cost, and disturbance sensitivity produce different orders.
- IBR-era disturbances make this operational: distorted waveforms and fast controls can turn a measurement artifact into a ride-through or forensic diagnosis problem.
- Estimator choice is conditional: event family, objective metric, and compute budget determine the preferred method.

Measurement rule

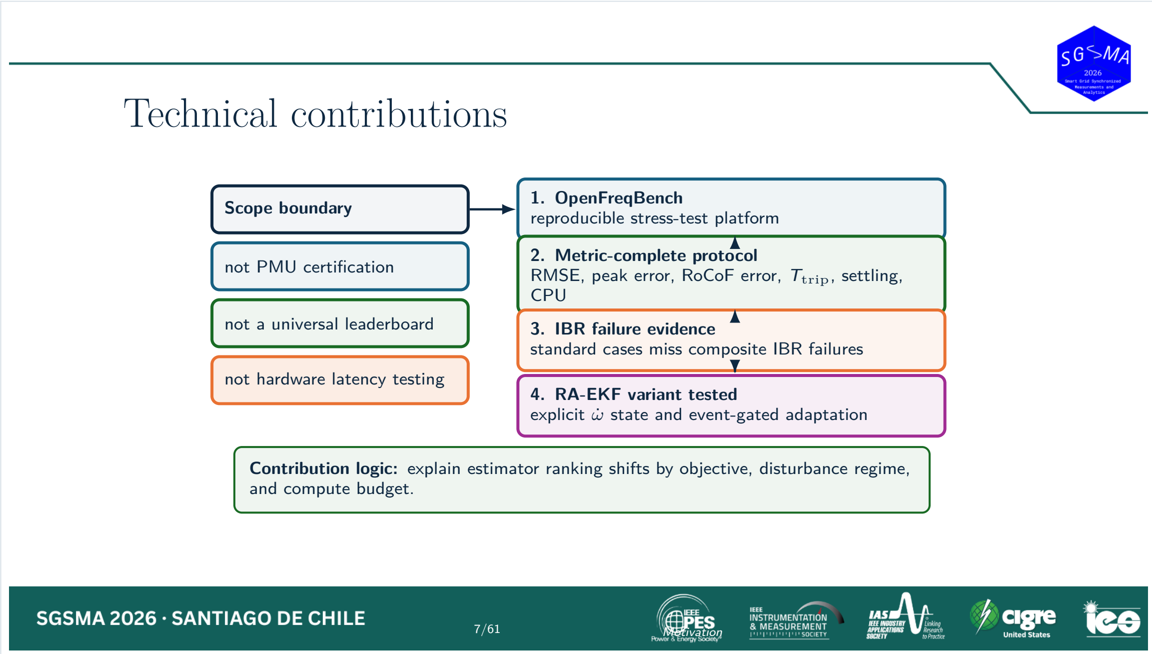
Report error, transient exposure, settling, latency, and cost as separate outputs for each estimator.

Operating claim

Purpose. Define what the work measures.

Speaker notes

- The claim is not "RA-EKF is always best" and not "one leaderboard solves it."
- Ranking changes when the objective changes: RMSE, RoCoF error, trip exposure, CPU cost, and recovery do not select the same estimator.
- Measurement rule: report error, transient exposure, setting, latency/cost, and outputs for each estimator.



Technical contributions

Purpose. Position the contribution cleanly.

Speaker notes

- Say what is not claimed: not device certification, not hardware latency measurement, not one universal leaderboard.
- Contributions: OpenFreqBench, metric-complete comparison, evidence that compliance-style tests miss IBR failure modes, and RA-EKF as one tested Kalman-family variant.
- Emphasize intellectual honesty: the benchmark explains when rankings change.

Experiment scale

16 estimators
loop, window, model

34 scenarios
standard + IBR

10 families
ramp to multi-event

60 MC runs
per pair

coverage
16 × 34 = **544**
summaries

replication
544 × 60 = **32,640**
records

sampling
1 MHz truth
10 kHz output

Experimental contract: fixed truth signals, scenarios, tuning policy, metrics, and timing before rankings are computed.

Central finding: no estimator dominates; winners change with event family, metric target, and compute budget.

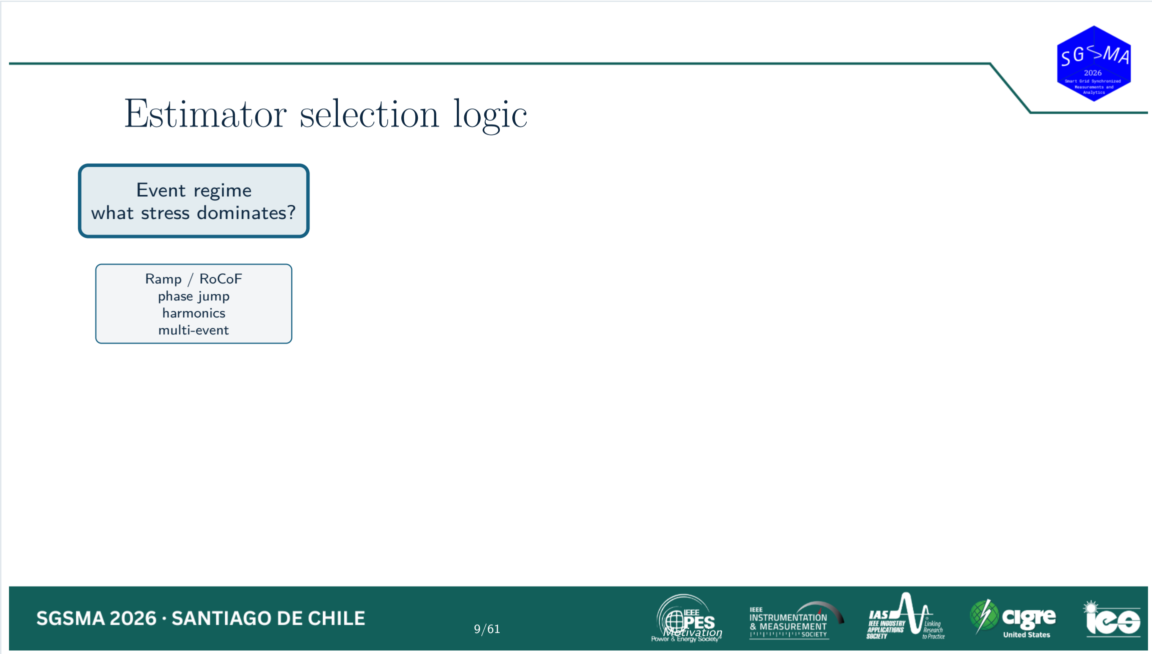
Source: SGSMA benchmark matrix, Sec. III-IV and Fig. 2 caption

Experiment scale

Purpose. Give the audience the scale early.

Speaker notes

- Keep evidence layers separate: the SGSMA benchmark matrix uses 16 estimators, 34 scenarios, and 60 Monte Carlo runs; the OpenFreqBench expansion uses 18 estimators, 33 scenarios, five seeds, and ATLAS sweeps.
- Explain that "grid search" means each method is given a fair chance within the tested parameter range.
- Key phrase: same truth signals, same I/O contract, same metrics.

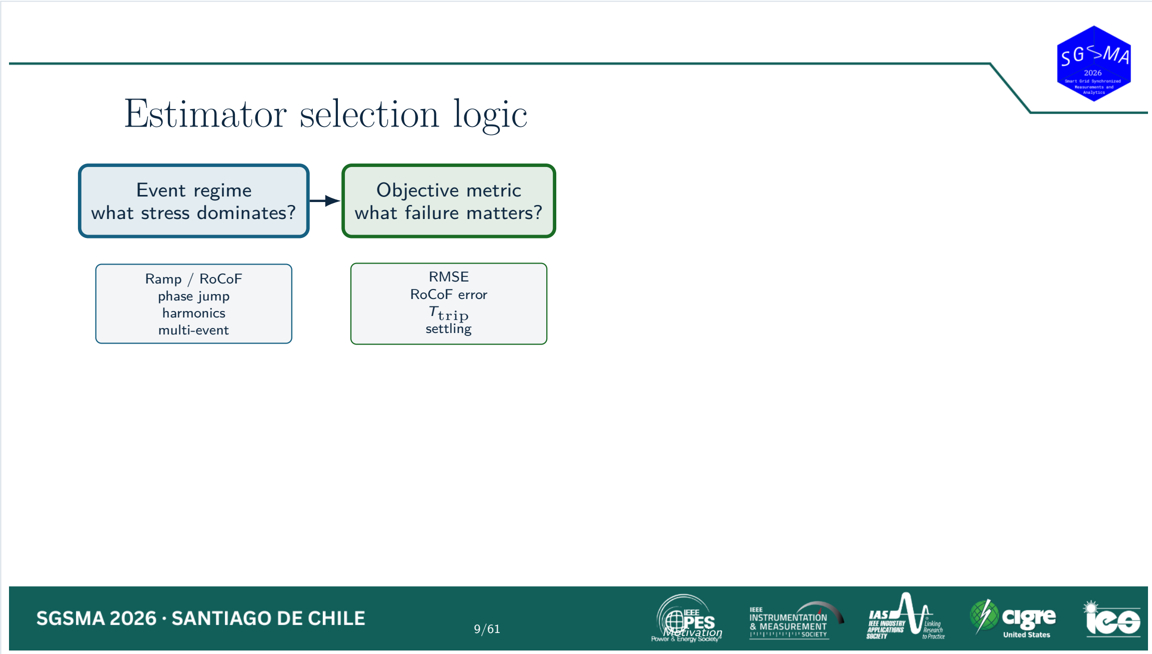


Selection logic: event family

Purpose. Start the decision map.

Speaker notes

- First question: what stress dominates the event? Ramp, phase jump, harmonic distortion, modulation, noise, or composite IBR dynamics.
- This prevents the audience from asking for one universal answer too early.
- Transition: after event family, choose the objective metric.

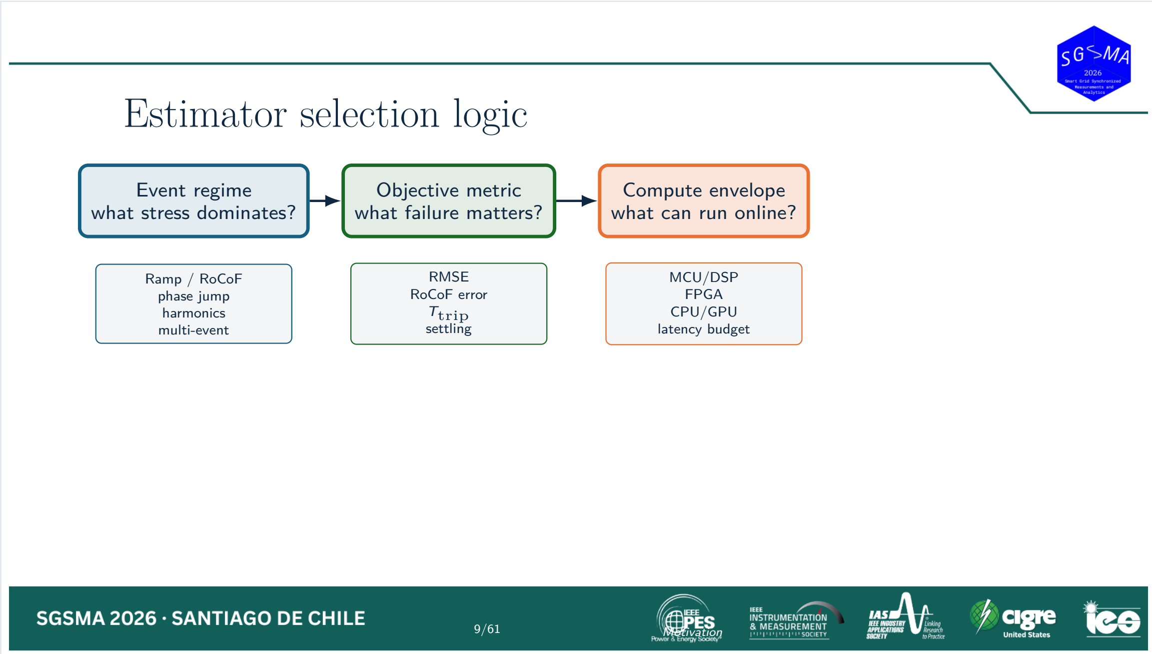


Selection logic: metric

Purpose. Explain why objectives matter.

Speaker notes

- The metric is not cosmetic. RMSE, RoCoF error, and trip-risk can produce different winners.
- Protection-adjacent use should care about duration above threshold and recovery, not just average error.
- Say: a method can be accurate on average and still unsafe during the transient.

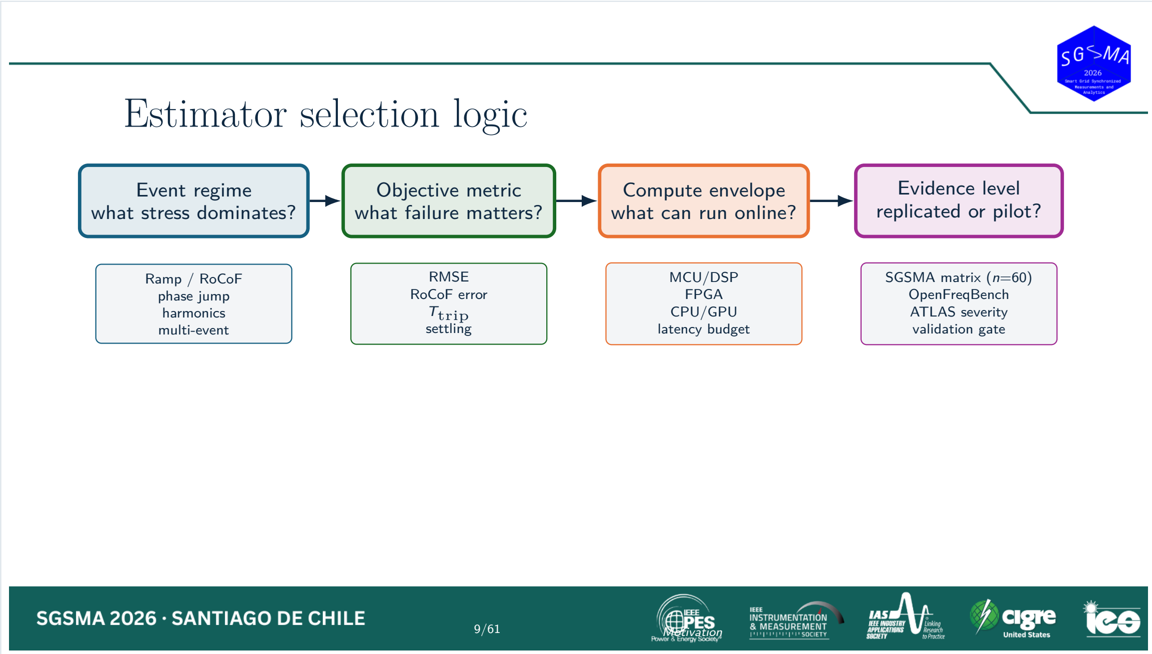


Selection logic: compute

Purpose. Add deployment constraints.

Speaker notes

- Compute cost changes the answer again.
- A high-accuracy method may be unacceptable for MCU/DSP or relay-adjacent screening if latency or CPU cost is too high.
- Keep this short; the Pareto slide will carry the evidence later.



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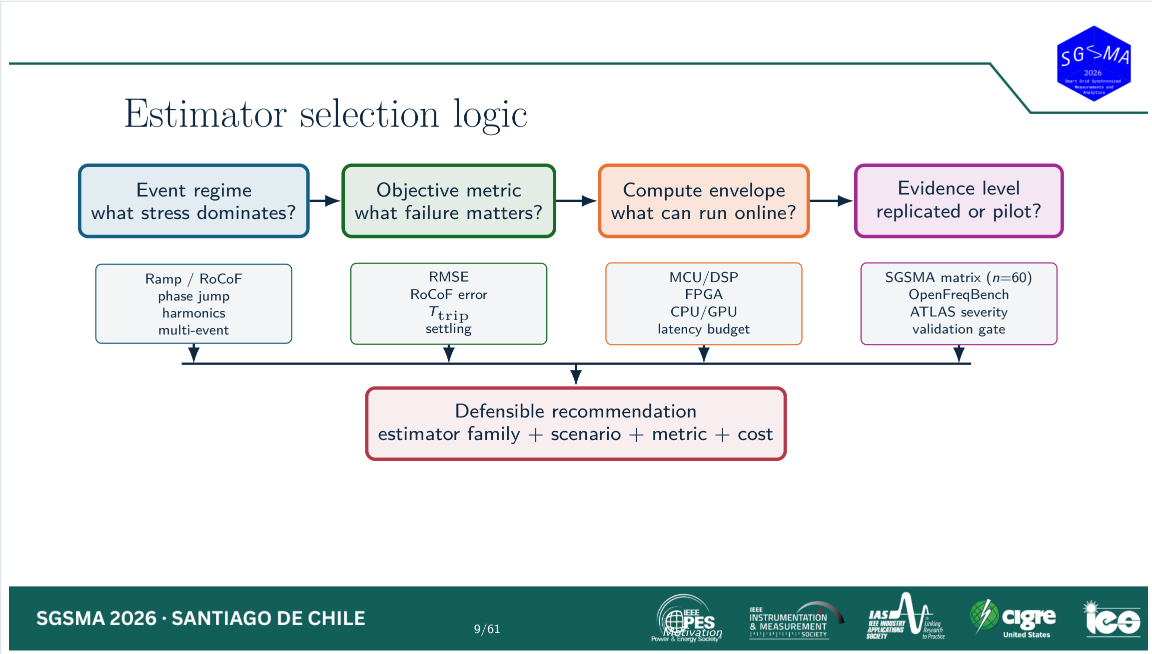
Selection logic: evidence level

Purpose. Separate pilot and validated evidence.

Speaker notes

- Evidence level matters. A pilot sweep can diagnose a failure mode; a validated replicated sweep supports a stronger claim.
- This is why ATLAS slides are labeled by sweep type.
- Avoid defensive language. Say: the labels keep the evidence contract explicit.



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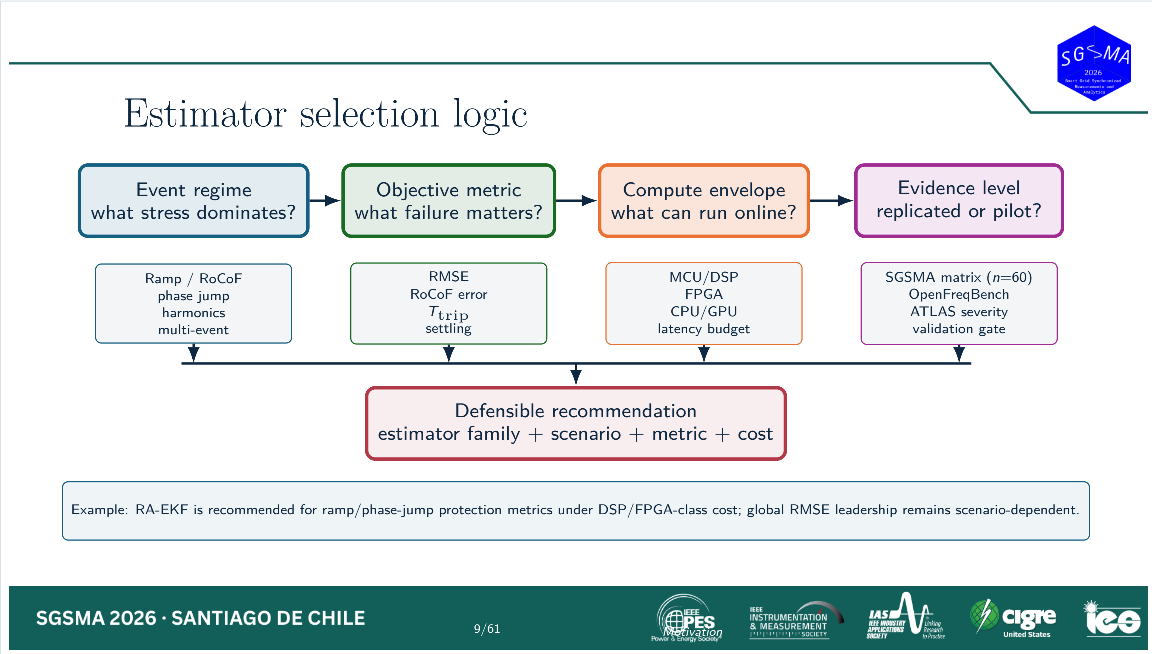
Selection logic: recommendation

Purpose. Close the logic chain.

Speaker notes

- The recommendation is conditional: estimator family plus scenario plus metric plus cost.
- Example: RA-EKF is defensible for ramp and phase-jump protection metrics under DSP/FPGA-like cost assumptions, but not as a global RMSE champion.
- This prepares the audience for the non-universal results.



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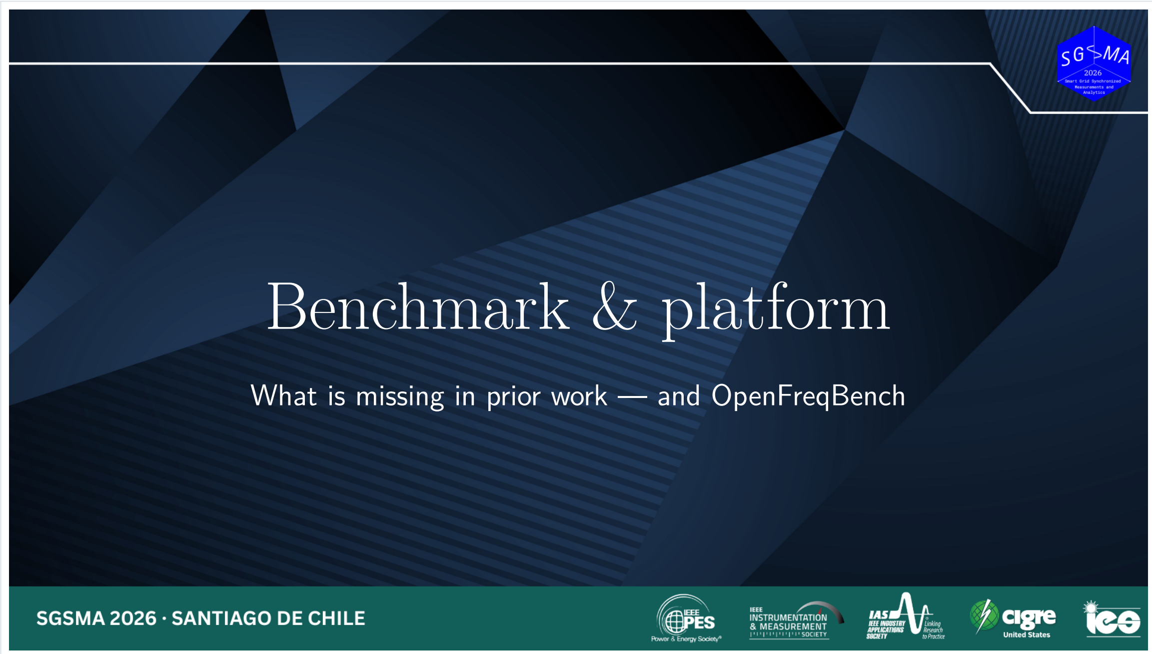
Selection logic: framing

Purpose. Reinforce the selection map.

Speaker notes

- Repeat the thesis once: the output is a defensible selection map, not a leaderboard.
- This is a good place to pause and let the framework settle.
- Transition to the benchmark platform.



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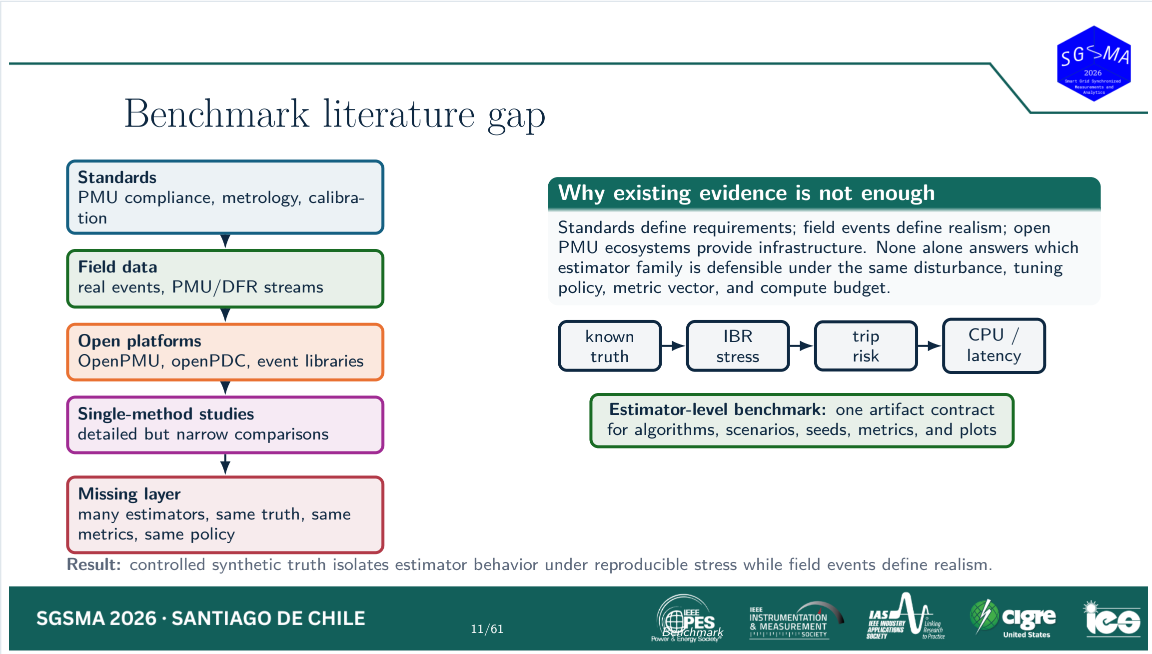
Benchmark and platform

Purpose. Introduce OpenFreqBench as infrastructure.

Speaker notes

- The platform is part of the contribution: it makes estimator comparison repeatable.
- The key idea is a public I/O contract, canonical scenarios, locked metrics, and machine-readable artifacts.
- Transition: before showing platform mechanics, show what previous evidence misses.



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Literature gap

Purpose. Explain why existing evidence is insufficient.

Speaker notes

- Standards define measurement requirements, field data defines realism, and open PMU ecosystems provide infrastructure.
- What is missing is estimator-level comparison under the same truth signals, stress cases, metrics, and policy.
- Say: controlled synthetic truth isolates estimator behavior; field events define the stress modes.

Benchmark comparison axes

Evidence source	Known truth	IBR composite stress	Trip-risk	CPU/latency	Reproducible artifacts
IEC/IEEE-style compliance	Yes	Limited	No	No	Device-dependent
Field PMU/DFR data	No	Yes	Possible	No	Dataset-dependent
Single-method studies	Often	Partial	Partial	Partial	Often limited
Estimator benchmark	Yes	Yes	Yes	Yes	Yes

Measured outputs

The same voltage trace is scored by RMSE, peak error, RoCoF error, trip exposure, settling, CPU, and structural latency.

Interpretive effect

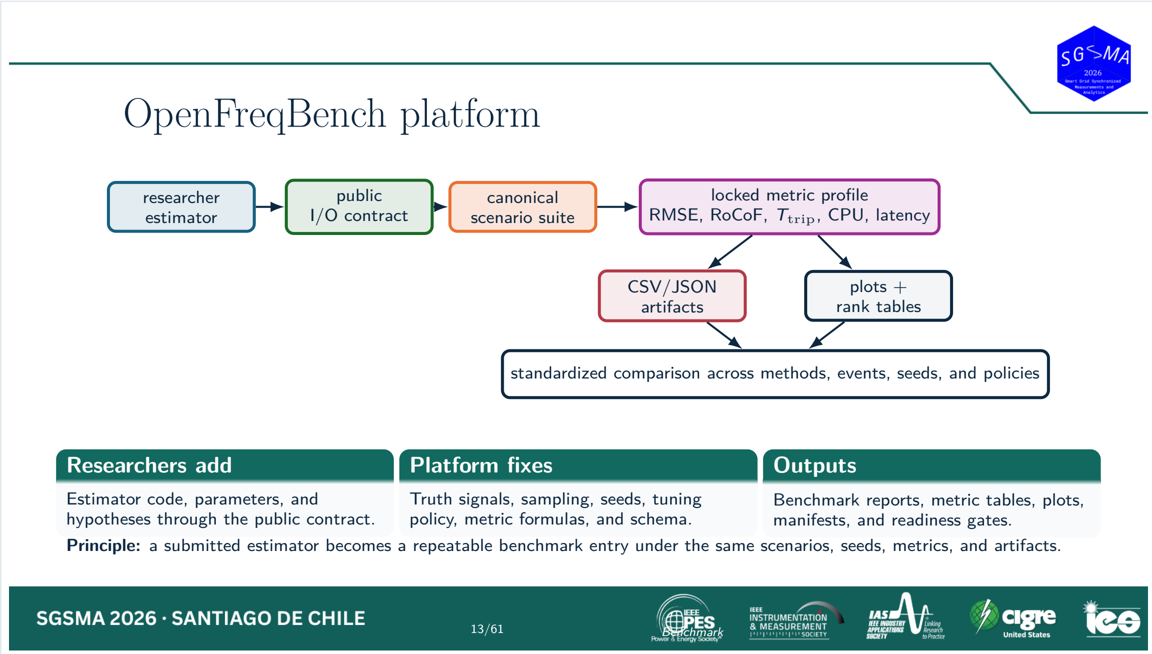
An estimator can pass standard cases and still fail a composite IBR event; it can also be accurate but too delayed or costly for protection-adjacent use.

Comparison axes

Purpose. Make the benchmark distinct from compliance testing.

Speaker notes

- Compliance tests and field records are both valuable, but they answer different questions.
- This benchmark tests algorithms, not devices, and reports trip-risk and compute as first-class outputs.
- Key line: passing a standard case can still fail to predict composite IBR readiness.



OpenFreqBench

Purpose. Make the platform memorable.

Speaker notes

- A researcher submits estimator code, parameters, and hypotheses through a public contract.
- The platform fixes truth signals, sampling, seeds, tuning policy, metric formulas, and schema.
- Key line: a new estimator becomes a repeatable benchmark entry, not a one-off comparison.

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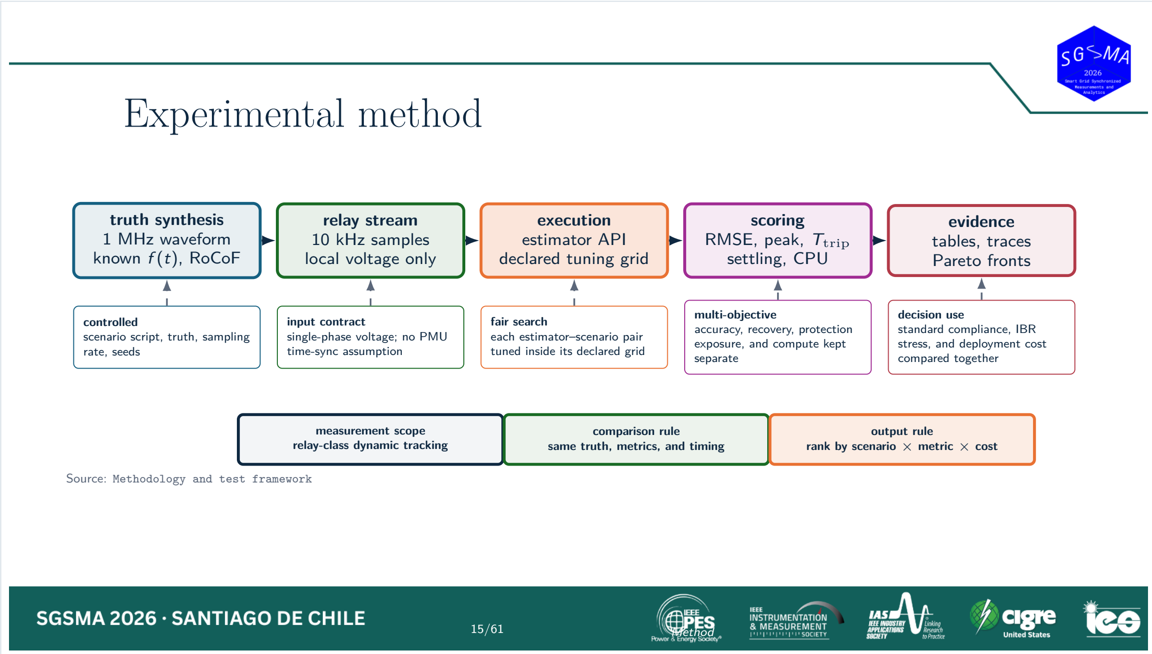
Experimental method divider

Purpose. Reset the audience.

Speaker notes

- This section answers "how was it tested?"
- Say: from here on, all comparisons use the same truth, same metrics, and fair tuning.
- Keep the transition short.



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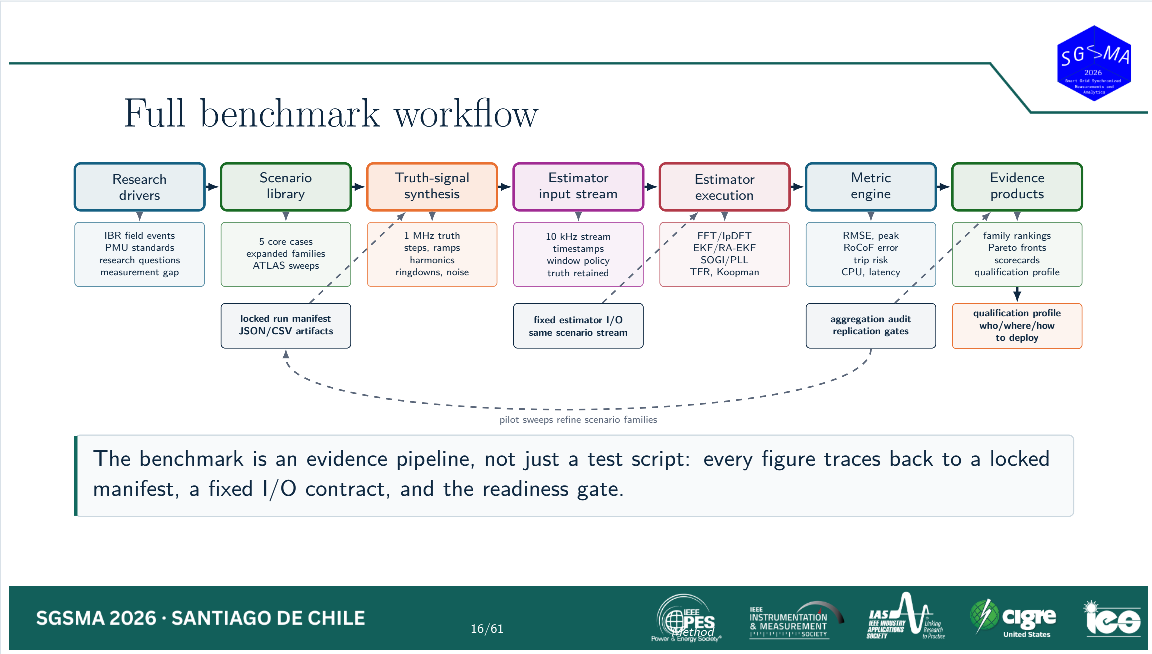
Experimental method

Purpose. Describe the data path.

Speaker notes

- Start with a 1 MHz waveform truth, then feed a 10 kHz DSP stream into estimator implementations.
- Each estimator-scenario pair is tuned within a declared grid and evaluated with RMSE, RoCoF, trip exposure, settling, and CPU.
- Clarify: this is not PMU timestamp certification; it is local non-time-synchronized relay-class estimation.



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Full workflow

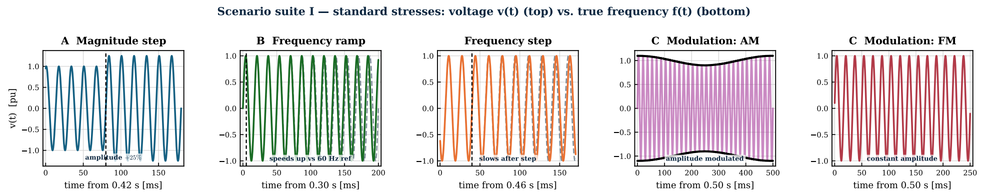
Purpose. Show traceability.

Speaker notes

- The workflow locks the research drivers, scenario library, truth-signal synthesis, estimator input stream, execution contract, metric engine, and evidence products.
- Emphasize reproducibility: every figure traces back to a manifest, I/O contract, and readiness gate.
- Do not spend too long here. The audience needs the map, not every box.

Standard stresses: the waveform is controlled

CORE SCENARIO SUITE



Each standard test changes one physical ingredient at a time: magnitude, ramp, frequency step, AM, or FM.

Standard waveforms

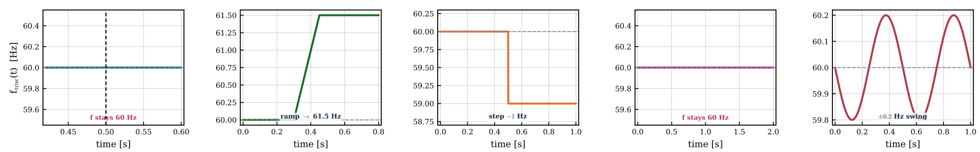
Purpose. Make the easy cases concrete.

Speaker notes

- These are controlled standard-style stresses: one physical ingredient changes at a time.
- The purpose is not to be realistic by itself; it is to isolate estimator behavior.
- Point to magnitude, ramp, frequency step, AM, and FM.

Standard stresses: the frequency truth is simple

CORE SCENARIO SUITE



The clean reference traces make estimator failure visible: divergence is an algorithmic response, not ambiguity in ground truth.

Standard frequency truth

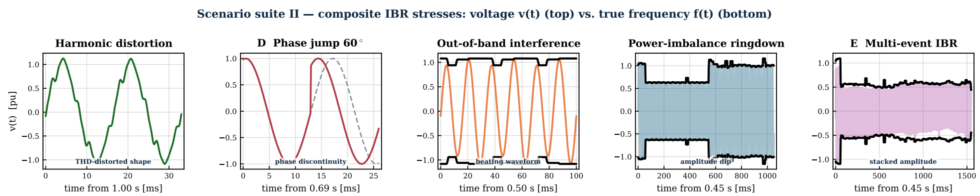
Purpose. Explain why failures are meaningful.

Speaker notes

- The frequency truth is simple and known.
- If an estimator diverges here, the failure is not ambiguity in the ground truth. It is algorithmic response.
- This sets up the ramp divergence result later.

Composite IBR stresses: waveform physics are stacked

IBR STRESS SUITE



Composite cases combine harmonic distortion, phase jumps, out-of-band tones, ringdown recovery, and multi-event amplitude dynamics.

Composite waveforms

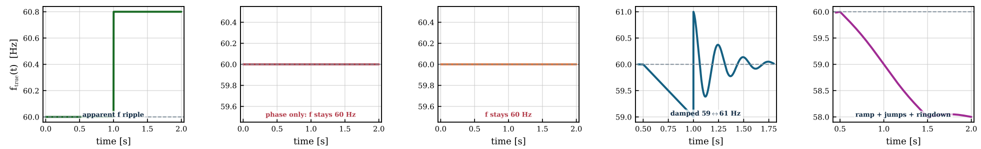
Purpose. Show why IBR cases are harder.

Speaker notes

- Composite cases stack physical stressors: harmonics, phase jumps, out-of-band tones, ringdown, and multi-event amplitude changes.
- These cases resemble the kind of mixed waveform conditions that break simple assumptions.
- Do not oversell realism; say they are controlled stress models motivated by field behavior.

Composite IBR stresses: frequency truth becomes diagnostic

IBR STRESS SUITE



The estimator must recover through nonstationary segments, not just report low steady-state error after the event.

Composite frequency truth

Purpose. Tie composite cases to recovery.

Speaker notes

- The estimator must recover through nonstationary segments.
- A low steady-state error after the event is not enough for protection or event reconstruction.
- Transition: now that scenarios are defined, compare estimator families.

Scenario families: distortion, leakage, and low SNR

Compact signal model

$$v(t) = A(t)\sin\theta(t) + v_h(t) + v_x(t) + n(t)$$

$$v_h(t) = \sum_{h=2}^{H} a_h \sin(h\theta + \phi_h), \quad v_x(t) = \sum_{q} b_q \sin(2\pi f_q t + \psi_q)$$

$$\mathrm{THD} = \frac{\sqrt{\sum_{h\geq 2} V_h^2}}{V_1}, \quad \mathrm{SNR}_{dB} = 20\log_{10}\left(\frac{1/\sqrt{2}}{\sigma_n}\right)$$

Interpretation

The benchmark keeps the truth signal and the waveform perturbation separate. When $f_{\mathrm{true}}(t)$ is fixed, estimator error measures artifact rejection, not a physical frequency event.

Family	What it tests
Harmonics	Integer multiples of 60 Hz; THD rejection with fixed true frequency.
Interharmonics	Non-integer tones; leakage and beating that can mimic dynamics.
OOB tones	Out-of-band interference; filter selectivity, aliasing margins, PLL leakage.
Noise / SNR	Controlled σ_n; low-SNR crossings and covariance assumptions.

Why this slide matters

In these cases $f_{\mathrm{true}}(t)$ can remain nominal. Any frequency excursion reported by the estimator is then a waveform-induced measurement artifact.

Scenario stress families

Purpose. Explain harmonics, interharmonics, OOB, and SNR before estimator families.

Speaker notes

- This slide defines the stress families that are easy to confuse: integer harmonics, non-integer interharmonics, out-of-band tones, and additive noise.
- The key distinction is that many of these cases keep the true frequency nominal. If the estimator moves, that is a measurement artifact caused by waveform distortion, leakage, or low SNR.
- Use the equations only as anchors: THD controls integer harmonic distortion; SNR is tied to the noise standard deviation relative to a 1 pu sine.

Estimator families

Loop-based

PLL, SOGI-PLL, SOGI-FLL, Type-3 SOGI-PLL.

Window / spectral

IpDFT, TFT, ESPRIT, Prony, MUSIC.

Model-based / stochastic

EKF, UKF, RA-EKF, LKF, LKF2.

Adaptive and data-driven

RLS, Koopman (RK-DPMU).

Legacy baselines

TKEO and ZCD are retained as comparison references outside the recommended deployment set.

Interpretation rule

These families occupy different latency and stress-sensitivity positions. The comparison tracks where each family wins, degrades, and fails.

LKF and LKF2 follow H. Ahmed, S. Biricik, M. Benbouzid, "Linear Kalman Filter-Based Grid Synchronization Technique: An Alternative Implementation," IEEE, 2021.

Estimator families

Purpose. Explain why family behavior matters.

Speaker notes

- Loop-based, spectral, state-space, adaptive, and legacy methods have different latency and stress sensitivities.
- Legacy baselines like ZCD and TKEO are kept as references, not recommended deployment solutions.
- The results will show where each family wins, degrades, and fails.

Metrics and decision variables

Error and recovery

$$e[k] = \widehat{f}[k] - f_{\mathrm{true}}[k]$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_k e[k]^2}$$

$$E_{\max} = \max_k |e[k]|$$

Protection-oriented risk

$$T_{\mathrm{trip}} = \Delta t \sum_k \mathbf{1}\{|e[k]| > 0.5\ \mathrm{Hz}\}$$

$$t_s = \min\{t : \ |e[\tau]| \le \epsilon,\ \forall \tau \ge t\}$$

$$C_{\mathrm{CPU}} = \text{mean time/sample}$$

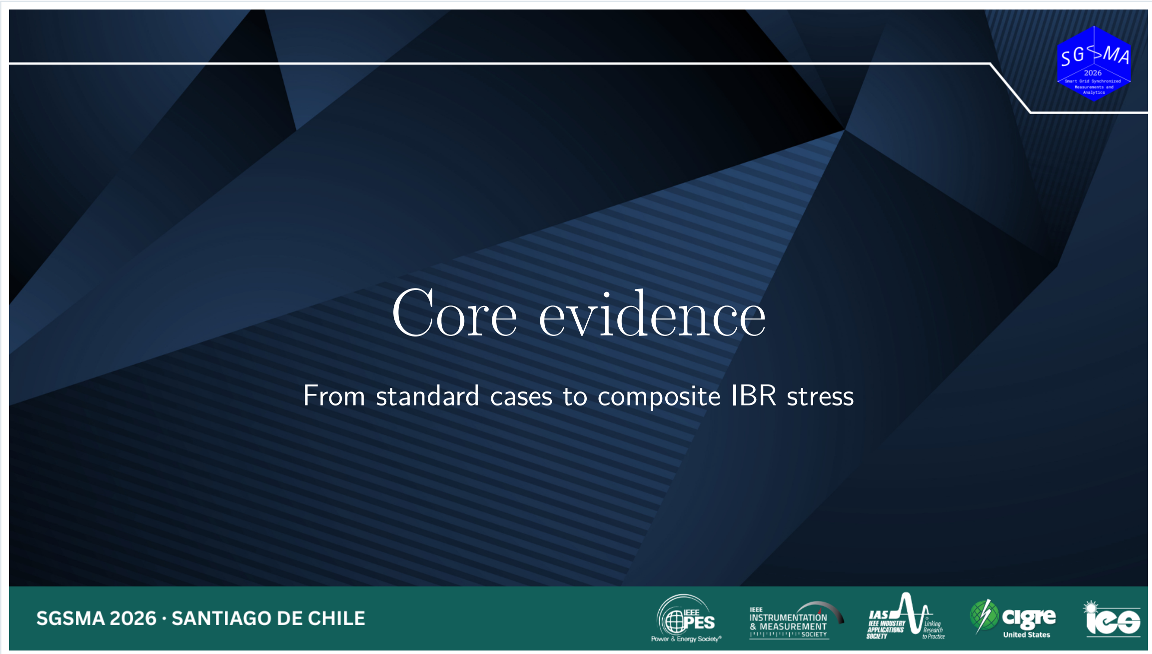
The 0.5 Hz threshold is a protection-oriented risk proxy. The metric set separates average error, transient severity, trip exposure, settling, and compute cost.

Metrics

Purpose. Define the scorecard.

Speaker notes

- RMSE measures average error; peak error and settling measure transient behavior.
- T_{trip} is a protection-oriented risk proxy: time spent above 0.5 Hz error.
- CPU cost is a software-level cost indicator, not hardware latency certification.



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Core evidence divider

Purpose. Shift from method to results.

Speaker notes

- Tell the audience: now we go from standard cases to composite IBR stress.
- The next slides establish the core empirical pattern.
- Pause briefly; this is the main evidence section.

Standard scenario results

Method	Step	Ramp	Mod.
RA-EKF	0.000	**0.006**	0.037
SOGI-FLL	0.005	0.013	0.040
Koopman	0.003	0.017	0.044
TFT	0.001	0.019	0.046
UKF	0.000	0.017	0.055
LKF2	0.138	0.015	**0.028**

RMSE [Hz], mean over 5 seeds. Green: below 0.05 Hz guide; red: failure/divergence regime.

What the heatmap shows

The standard step is easy for most methods. The ramp is the separator: EKF and IpDFT can look excellent at a point and still diverge in another seed.

Operational consequence

Robustness is not the same as point accuracy. A standard scenario can pass while the same estimator remains unsafe for ramp-like IBR dynamics.

Standard results: stable methods

Purpose. Read the heatmap first.

Speaker notes

- Start with the green rows: RA-EKF, SOGI-FLL, Koopman, and TFT remain under the 0.05 Hz guide across the standard cases.
- The point is not that all methods fail. Some are strong under clean controlled tests.
- Then prepare the reveal: clean cases can still hide divergence.

Standard scenario results

Method	Step	Ramp	Mod.
RA-EKF	0.000	**0.006**	0.037
SOGI-FLL	0.005	0.013	0.040
Koopman	0.003	0.017	0.044
TFT	0.001	0.019	0.046
UKF	0.000	0.017	0.055
LKF2	0.138	0.015	**0.028**
PLL	0.033	**1.47**	**0.227**
IpDFT	0.000	**14.4**	0.114
EKF	0.000	**37.1**	0.064

RMSE [Hz], mean over 5 seeds. Green: below 0.05 Hz guide; red: failure/divergence regime.

What the heatmap shows

The standard step is easy for most methods. The ramp is the separator: EKF and IpDFT can look excellent at a point and still diverge in another seed.

Operational consequence

Robustness is not the same as point accuracy. A standard scenario can pass while the same estimator remains unsafe for ramp-like IBR dynamics.

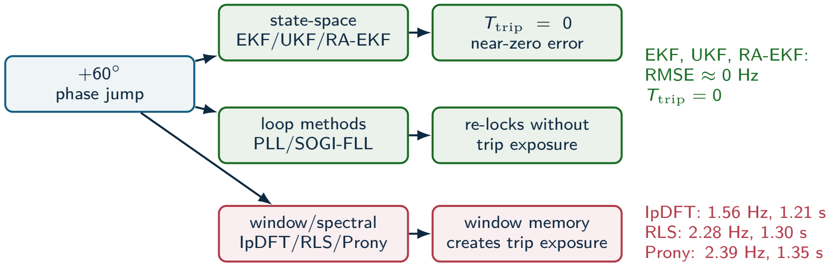
Standard results: divergence appears

Purpose. Explain the red cells.

Speaker notes

- The ramp separates methods. EKF and IpDFT can look excellent at one point and diverge in another seed.
- Main line: robustness is not the same as point accuracy.
- Transition to phase jumps, where the family pattern changes again.

A clean phase jump is a family separator



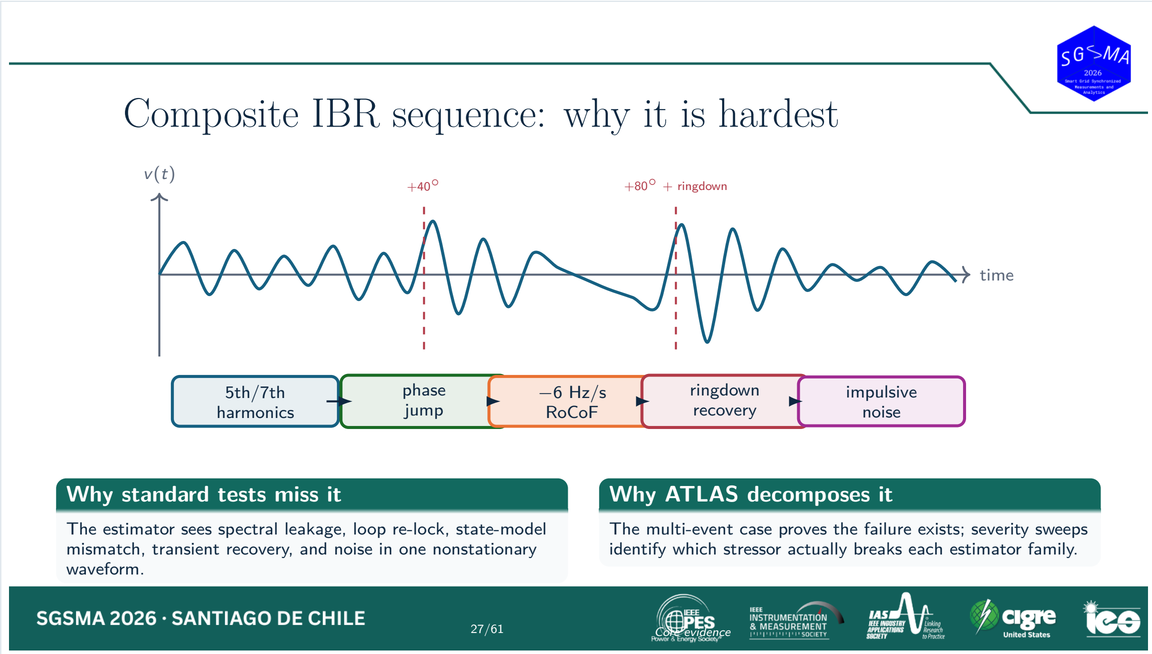
The clean jump does not expose RA-EKF's main advantage. It shows which estimator families carry memory through discontinuities.

Phase jump family separator

Purpose. Use the mechanism diagram.

Speaker notes

- A clean 60 degree phase jump does not break state-space filters or loop methods.
- Windowed and spectral methods carry memory through the discontinuity, creating trip exposure.
- Important nuance: RA-EKF's strongest value is not this clean phase jump; it appears in ramp and multi-event stress.



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Composite IBR sequence

Purpose. Explain why this is hardest.

Speaker notes

- The multi-event trace stacks harmonics, phase jumps, RoCoF, ringdown recovery, and noise.
- Standard tests miss this because each stressor is isolated.
- ATLAS decomposes the composite case to find which stressor breaks each estimator family.

Scenario E: the leaderboard breaks

Method	RMSE [Hz]	T_{trip} [s]	CPU [μs]
UKF	**1.09**	**0.78**	14
EKF	1.13	0.87	11
SOGI-PLL	1.14	**0.78**	11
SOGI-FLL	1.14	0.84	**9**
IpDFT	1.14	1.35	17
RA-EKF	1.16	0.92	15
Koopman	1.90	0.84	**15 241**
ZCD	**426**	1.55	7

No accurate winner

Under the stacked sequence every usable method collapses to $\approx$ 1.1 Hz RMSE; ZCD diverges catastrophically (426 Hz). High accuracy is simply not available here.

Scalar ranking failure

UKF has the lowest RMSE and trip exposure, SOGI-FLL the lowest usable CPU, Koopman is 1000$\times$ costlier; RA-EKF sits mid-pack but never diverges.

Source: `full_mc_benchmark`, $n = 5$: `IBR_Multi_Event`

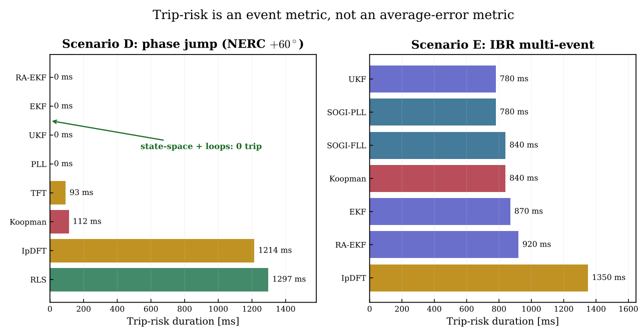
Scenario E leaderboard breaks

Purpose. Make the result uncomfortable.

Speaker notes

- Under stacked IBR stress, every usable method collapses to about 1.1 Hz RMSE; ZCD diverges catastrophically.
- UKF has the lowest RMSE and trip exposure, SOGI-FLL has the lowest usable CPU, and Koopman is much more costly.
- Key line: a scalar ranking fails because the objectives disagree.

Trip-risk evidence: core stress cases



Result reading

- Scenario D: state-space filters and PLL keep $T_{\mathrm{trip}} = 0$; windowed/spectral methods (IpDFT, RLS) exceed 1.2 s.
- Scenario E: *every* method carries 0.8–1.4 s exposure — composite stress is unavoidable.
- The trip-exposure ranking is not the RMSE ranking.

Result: T_{trip} separates protection exposure from average error.

Trip-risk evidence

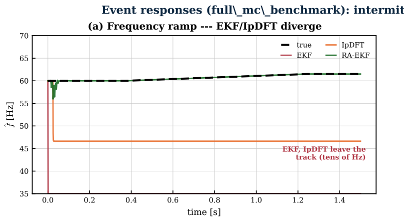
Purpose. Separate trip exposure from RMSE.

Speaker notes

- Scenario D shows zero trip exposure for state-space and loop methods, but long exposure for windowed methods.
- Scenario E shows nearly every method carries significant exposure.
- Main point: T_{trip} answers a protection question that RMSE does not.

The failures are visible in the trace, not only in the table

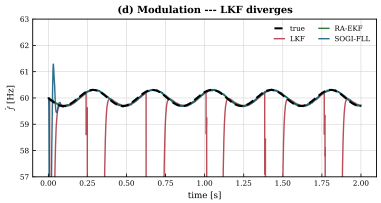
RAMP DIVERGENCE



Failure signature

EKF and IpDFT leave the true ramp almost immediately; after the event, the estimate is no longer operationally usable.

MODULATION DIVERGENCE



Failure signature

LKF stays plausible between collapses, then repeatedly drops out of the usable frequency band.

Trace-visible failures

Purpose. Use the plots as evidence.

Speaker notes

- Left: EKF and IpDFT leave the ramp almost immediately; the estimate is not operationally usable.
- Right: LKF looks plausible between collapses, then repeatedly drops out.
- Say: this is why trace-level evidence is needed, not only tables.

Result comparison: winners depend on the question

Question	Best-supported answer	Evidence	Boundary
Clean step/modulation accuracy	State-space + SOGI-FLL near-zero; LKF2 best on mod	Step ≈ 0; mod 0.028–0.044 Hz	Clean cases do not predict ramp/event survival
Ramp and RoCoF tracking	RA-EKF is the most robust ramp tracker	0.006 Hz vs EKF 37 Hz, IpDFT 14 Hz (diverge)	Divergence is intermittent (seed-dependent)
Phase jump (Scenario D)	State-space filters recover with zero trip	EKF/UKF/RA-EKF $T_{\mathrm{trip}} = 0$; IpDFT 1.21 s	Windowed/spectral carry the exposure
Multi-event RMSE	No accurate method — all ≈ 1.1 Hz	UKF 1.09 Hz best; ZCD 426 Hz diverges	RMSE hides trip-risk and 1000× CPU spread
Embedded feasibility	PLL, SOGI-FLL, EKF, RA-EKF remain plausible	9–17 μs/sample; Koopman ∼15,000	Python timing is a software-cost proxy

Selection changes with event family, objective metric, and compute envelope. A single leaderboard hides the deployment trade-off.

Question-dependent winners

Purpose. Summarize the core table.

Speaker notes

- The best-supported answer depends on the question.
- Clean step/modulation, ramp tracking, phase jump recovery, multi-event RMSE, and embedded feasibility all favor different methods.
- This slide is the bridge from data to deployment.

Computational feasibility

Method	Time/sample	Family	Target
SOGI-FLL	0.0348 μs	Loop	MCU
SOGI-PLL	0.0867 μs	Loop	MCU
EKF	0.813 μs	Model	MCU/DSP
RA-EKF	**1.61 μs**	**Model**	**DSP**
TFT / IpDFT	3–4 μs	Window	DSP/FPGA
Koopman	390 μs	Data-driven	CPU/GPU
ESPRIT	2,838 μs	Window	CPU

Source: SGSMA benchmark matrix, Sec. IV-D and Fig. 2(g)-(h)

Deployment interpretation

CPU is a software-cost indicator from the benchmark harness, not a hardware-latency claim.

Key comparison

The practical low-cost cluster is SOGI-FLL, SOGI-PLL, EKF, and RA-EKF. Koopman and ESPRIT sit far to the right on cost without a global RMSE gain.

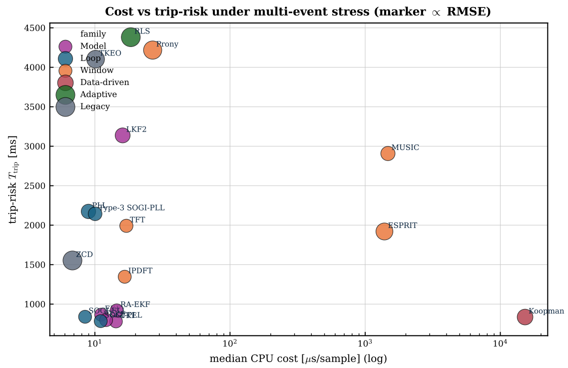
Computational feasibility

Purpose. Clarify CPU interpretation.

Speaker notes

- Python timing separates estimator families but is not hardware latency.
- Loop and state-space methods are low-cost; spectral and data-driven methods can be much more expensive.
- State the limitation clearly: hardware timing requires HIL or platform measurements.

Deployment limits



Source: OpenFreqBench expansion: artifacts/full_mc_benchmark, $n = 5$

Deployment interpretation

- Deployment combines RMSE, trip exposure, CPU, and divergence robustness.
- In the OpenFreqBench expansion, state-space + loop methods cluster at low cost ($\leq 17\,\mu$s) and low trip.
- Spectral/data-driven methods trade lower error for much higher CPU in this artifact.
- RA-EKF stays low-cost with no ramp divergence in this run.

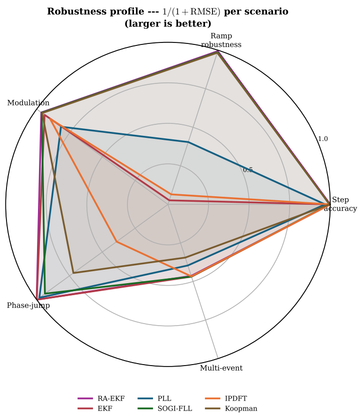
Deployment limits

Purpose. Read the Pareto structure.

Speaker notes

- Deployment combines RMSE, trip exposure, CPU, and divergence.
- The low-cost cluster is not automatically best; the expensive cluster is not automatically deployable.
- RA-EKF stays low-cost with no ramp divergence in this run.

Balance profile



Radar profile

- Each axis is one scenario; larger = more robust ($1/(1 + \mathrm{RMSE})$).
- RA-EKF and SOGI-FLL stay robust across all five scenarios.
- EKF and IpDFT collapse on the ramp axis (intermittent divergence).
- No estimator is strong on every axis — robustness is the discriminator.

Balance profile

Purpose. Use the radar as a quick synthesis.

Speaker notes

- Each axis is one scenario; larger means more robust.
- RA-EKF and SOGI-FLL remain robust across the five core scenarios.
- No estimator is strong everywhere. Robustness is the discriminator.

Compliance heatmap: why standard tests are insufficient

Method	Step	Ramp	Mod	Jump	Multi
Koopman	P	P	P		
TFT	P	P	P		
IpDFT	P	F	P		
SOGI-FLL	P	P	P		
PLL	P	F	F*		
EKF	P	F	P		
UKF	P	P	P		
RA-EKF	P	P	P		
LKF2	F*	P	P		

Columns: **Jump** = phase jump (D), **Multi** = multi-event (E). **P** pass | **F** fails a declared RMSE/T_{trip}/settling threshold | **F*** borderline.

Compliance heatmap: first read

Purpose. Show the standard-test limitation.

Speaker notes

- Many methods pass step, ramp, and modulation columns.
- Do not conclude readiness from those passes.
- Wait for the jump and multi-event columns.

Compliance heatmap: why standard tests are insufficient

Method	Step	Ramp	Mod	Jump	Multi
Koopman	P	P	P	F	F
TFT	P	P	P	F*	F
IpDFT	P	F	P	F	F
SOGI-FLL	P	P	P	P	F
PLL	P	F	F*	P	F
EKF	P	F	P	P	F
UKF	P	P	P	P	F
RA-EKF	P	P	P	P	F
LKF2	F*	P	P	P	F

Columns: **Jump** = phase jump (D), **Multi** = multi-event (E). **P** pass | **F** fails a declared RMSE/T_{trip}/settling threshold | **F*** borderline.

Result: Step, ramp, and modulation compliance does not predict phase-jump or multi-event readiness.

Compliance heatmap: failure read

Purpose. State the punchline.

Speaker notes

- The jump and multi-event columns show failures that standard cases did not predict.
- Passing clean cases is necessary but not sufficient for IBR readiness.
- Key line: standard tests do not predict composite event survival.

Empirical claims

Protection behavior

Divergence, not phase jumps, is the dominant failure mode.
On the ramp, EKF and IpDFT intermittently diverge (37 and 14 Hz mean); RA-EKF stays at 0.006 Hz. A clean phase jump is recovered by every state-space filter with $T_{\mathrm{trip}} = 0$.

Metric disagreement

RMSE, trip-risk, and CPU select different winners.
In the OpenFreqBench multi-event run, UKF has the lowest RMSE/trip, SOGI-FLL the lowest usable CPU, and Koopman is 1000× costlier.

Standard-test limitation

Simple-case pass/fail leaves event readiness unresolved.
The heatmap shows methods that pass Step/Ramp/Mod but fail islanding or multi-event IBR stress.

Deployability

Feasibility changes the accuracy ranking.
In that same expansion, RA-EKF runs at $\sim$15 μs/sample; selected spectral and data-driven methods move far outside that cost range.

Guardrail: report scenario, metric, and compute envelope before naming a preferred estimator.

Source: OpenFreqBench expansion: `artifacts/full_mc_benchmark`, $n = 5$

Empirical claims

Purpose. Consolidate the core evidence.

Speaker notes

- Four claims: protection behavior, metric disagreement, standard-test limitation, and deployability.
- Keep it crisp. This is not a new result slide; it is a synthesis slide.
- Transition: next section asks whether the pattern scales across families and severity.

Click/page 44 of 74 Source: `slide.pdf`

Expanded and ATLAS divider

Purpose. Open the extended evidence.

Speaker notes

- Tell the audience: now we move from core benchmark cases to severity sweeps.
- The purpose of ATLAS is diagnosis: find where and why estimator behavior changes.
- Keep the evidence-level labels visible.

Expanded stress analysis

OPENFREQBENCH EXPANSION: 18 estimators, 33 scenarios, $n = 5$ seeds per scenario

Monte-Carlo benchmark (full_mc)

- 18 estimators across 5 families.
- 33 scenarios (A–E plus severity variants).
- $n = 5$ Monte Carlo seeds per scenario.
- Standard, phase-jump, multi-event, and CPU metrics.

ATLAS severity sweeps

- Severity, sign, and policy curves per stress family.
- Fixed-policy $n = 30$ across all 9 required sweeps (both signs).
- Harmonic, RoCoF, phase-jump, and noise atlases.
- Tests sensitivity, scaling, and family stability.

Two evidence layers on the same truth: a replicated Monte-Carlo benchmark, plus ATLAS severity sweeps that trace where each estimator family degrades.

Expanded stress analysis

Purpose. Explain the two evidence layers.

Speaker notes

- Full Monte Carlo benchmark gives replicated scenario rankings.
- ATLAS severity sweeps map thresholds, signs, and policy sensitivity.
- Key line: one layer ranks methods; the other explains where the ranking breaks.

SGSMA benchmark: family winners

Family	Count	RMSE leader	RMSE mean	Trip-risk leader
IEEE ramps	9	SOGI-PLL	19.4 ± 6.4 mHz	0 s (multiple)
Magnitude steps	7	EKF	0.388 ± 0.143 mHz	0 s (multiple)
Modulation	3	RA-EKF	17.8 ± 21.4 mHz	0 s (multiple)
Phase jumps	3	SOGI-FLL	174 ± 67.5 mHz	ZCD: 0.0194 s
OOB interference	1	EKF	13.6 ± 11.1 mHz	EKF / RA-EKF: 0 s
IBR harmonics	3	EKF	70.5 ± 27.3 mHz	TFT / Prony: 0.00947 s
IBR ringdown	5	UKF	313 ± 23.2 mHz	ESPRIT: 0.0989 s
IBR multi-event	1	PLL	173.5 ± 66.7 mHz	IpDFT: 0.0439 s

The paper result is conditional: SOGI-PLL leads ramps, EKF leads magnitude steps/OOB/harmonics, RA-EKF leads modulation, UKF leads ringdown, and PLL leads the composite IBR case. Trip-risk often points elsewhere.

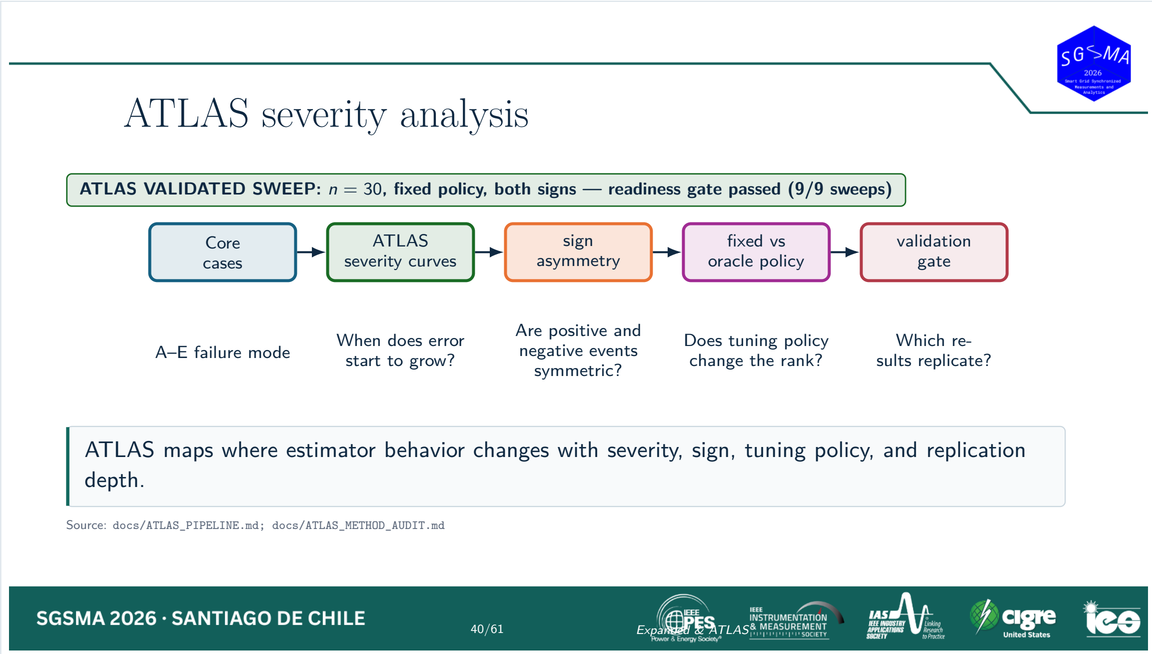
Source: SGSMA benchmark matrix, Table II

Family winners

Purpose. State the no-universal-winner pattern.

Speaker notes

- The family winners rotate across event types.
- RA-EKF, UKF, LKF2, SOGI-FLL, and ESPRIT all win somewhere.
- The multi-event case remains the hardest: error grows from sub-mHz clean cases to about 1 Hz.



Click/page 47 of 74 Source: `slide.pdf`

ATLAS method

Purpose. Explain the readiness pipeline.

Speaker notes

- ATLAS asks: when does error grow, are signs asymmetric, does fixed policy change the rank, and which result replicates?
- This is a diagnostic microscope, not just another leaderboard.
- It supports stress-specific qualification.

ATLAS result: metric choice changes the answer

ATLAS VALIDATED SWEEP: $n = 30$, **fixed policy, both signs — readiness gate passed (9/9 sweeps)**

Metric objective	Unique metric winner	Different from RMSE
Trip-risk exposure	26 / 113	5 / 26
RoCoF max error	112 / 113	53 / 112
RoCoF RMS error	112 / 113	48 / 112

Why this matters

For protection-adjacent use, low average error is not enough. The estimator must avoid long deadband exposure, track RoCoF, and remain computable.

Tie handling

Trip-risk has tied zero-exposure minima in 87/113 levels. It is an exposure screen first, not a single-winner leaderboard.

Operational implication

RoCoF metrics change the preferred estimator in nearly half the severity map; trip-risk removes estimators with long exposure.

Source: `atlas-papergrade-missing-v1/global_metrics_report.csv`

Metric choice changes the answer

Purpose. Make this a headline result.

Speaker notes

- Across 113 severity levels, RoCoF metrics frequently change the leader.
- Trip-risk must be read tie-aware: 87 levels have tied zero-exposure minima, so it is an exposure screen before it is a leaderboard.
- Key line: the benchmark output is a metric-conditioned selection map.

ATLAS result: where estimators stop being valid

ATLAS VALIDATED SWEEP: $n = 30$, fixed policy, both signs — readiness gate passed (9/9 sweeps)

Stress family	Maximum tested level	Methods below 0.05 Hz	Interpretation
Magnitude step	±90% sag/swell	2	EKF and RA-EKF survive both signs.
AM modulation	10 Hz envelope	10	AM alone is not the hardest modulation case.
FM modulation	10 Hz modulation	0	FM is qualitatively harder than AM.
RoCoF ramp	±50 Hz/s	0	Severe RoCoF requires a separate qualification class.
Phase jump	$\pm 60^\circ$	0	Single jump severity already breaks the 0.05 Hz guide.
Harmonics	30% THD	0	Low-THD winners do not survive extreme THD.
Interharmonics	20% at 75 Hz	0	Integer-harmonic tests are not sufficient.
Broadband noise	$\sigma = 0.1$ pu	0	Noise robustness is a first-class stress dimension.

Where estimators stop being valid

Purpose. Read the threshold table.

Speaker notes

- Severe FM, RoCoF, phase jumps, interharmonics, high THD, and broadband noise leave no method below the 0.05 Hz guide at the maximum tested level.
- Magnitude sag/swell is different: EKF and RA-EKF survive both signs.
- This slide supports qualification by stress family.

AM, FM, and interharmonics split the story

ATLAS VALIDATED SWEEP: $n = 30$, **fixed policy, both signs — readiness gate passed (9/9 sweeps)**

Stress	Winner pattern	Threshold behavior
AM modulation	EKF wins slow, reference, fast, and severe regions.	Ten methods stay below 0.05 Hz through 10 Hz envelope modulation.
FM modulation	SOGI-FLL, TFT, and Koopman split the low-rate points; ESPRIT leads from 1–10 Hz.	No method stays below 0.05 Hz at 10 Hz FM modulation.
Interharmonics	EKF leads 0.5–15%; PLL leads at 20%.	EKF crosses the 0.05 Hz guide between 5% and 8%; most families cross earlier.

Implication

AM, FM, integer harmonics, and interharmonics should not be collapsed into one "distortion" label. They select different estimator families.

Decomposition insight

A composite IBR trace can contain AM, FM, harmonics, interharmonics, noise, and phase steps. ATLAS isolates which stressor changes the ranking.

AM, FM, interharmonics

Purpose. Separate distortion types.

Speaker notes

- AM, FM, harmonics, and interharmonics should not be collapsed into one distortion label.
- AM remains easier; FM and interharmonics select different estimator families.
- Composite IBR traces contain several of these at once, so decomposition matters.

When the textbook filter diverges

OPENFREQBENCH EXPANSION: 18 estimators, 33 scenarios, $n = 5$ seeds per scenario

Per-seed frequency RMSE [Hz] on the IEEE frequency-ramp scenario (5 Monte Carlo seeds):

Estimator	seed 1	seed 2	seed 3	seed 4	seed 5	trip-risk
EKF (textbook)	0.016	63.7	0.017	121.5	0.027	up to 1.35 s
RA-EKF	0.0055	0.0062	0.0083	0.0045	0.0058	0 s

What happens

The standard EKF tracks most seeds but **intermittently diverges** to tens of Hz. It is *not* a severity threshold: the worst seed (121 Hz) had a low ramp rate (1.7 Hz/s) while the steepest (5 Hz/s) tracked cleanly. The failure is triggered by onset/phase combinations — seed-dependent and silent.

Why RA-EKF matters here

Covariance scaling and event gating remove the divergence: stable on **every** seed, zero trip exposure. This — not a universal RMSE win — is its defensible value.

Textbook EKF divergence

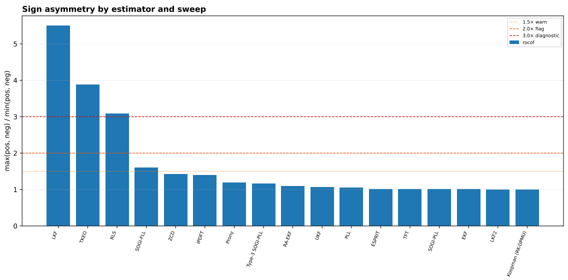
Purpose. Explain the RA-EKF value.

Speaker notes

- Standard EKF tracks most seeds but silently diverges on some onset/phase combinations.
- RA-EKF stays stable on every seed and keeps zero trip exposure.
- Do not claim universal superiority. Claim robust ramp behavior under these tested conditions.

RoCoF has a sign: several methods are down-ramp sensitive

ATLAS VALIDATED SWEEP: $n = 30$**, fixed policy, both signs — readiness gate passed (9/9 sweeps)**



Source: atlas-papergrade-missing-v1/atlas_sign_asymmetry.csv ($n = 30$, both signs)

Directional bias (RMSE$_-$/RMSE$_+$ at $\pm$50 Hz/s)

- LKF **5.5**$\times$ worse on $-$RoCoF
- TKEO **3.9**$\times$, RLS **3.1**$\times$
- SOGI-FLL 1.6$\times$
- EKF, UKF, RA-EKF $\approx$ 1.0 (symmetric)

Why it matters

For LKF, TKEO, RLS, and SOGI-FLL, a $+$RoCoF-only test can certify behavior that does not hold on frequency *decline*. Both signs must be swept.

RoCoF sign asymmetry

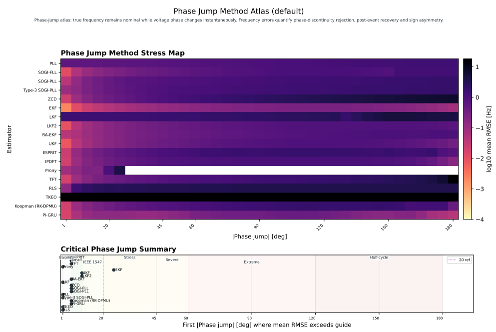
Purpose. Make the sign result memorable.

Speaker notes

- Down-ramps are harder for several methods: LKF, TKEO, RLS, and SOGI-FLL show directional bias.
- EKF, UKF, and RA-EKF are approximately symmetric at this tested level.
- Key line: a positive-only RoCoF test can certify a method that fails in the protection-critical decline direction.

Phase-jump severity is non-monotone

ATLAS SEVERITY SWEEP: pilot run, $n = 1$, default controller policy



Worst case is not 180°

Most methods peak near 90–120° and *recover* toward 180° (a half-cycle jump partly self-cancels). EKF peaks at only **0.25 Hz @ 115°**, back to 0.07 Hz @ 180°.

The TFT cliff

One exception: TFT rises gently, then jumps **2.5 → 21.9 Hz** between 150° and 180° — a phase-reversal singularity.

Result: Severity is non-monotone; sweep the jump angle rather than qualify one nominal point.

Source: `atlas-phase-jump-1-180deg-5deg-all18/global_metrics_report.csv` (diagnostic, $n = 1$, default policy, positive sign)

Phase-jump non-monotonicity

Purpose. Explain why one nominal point is weak.

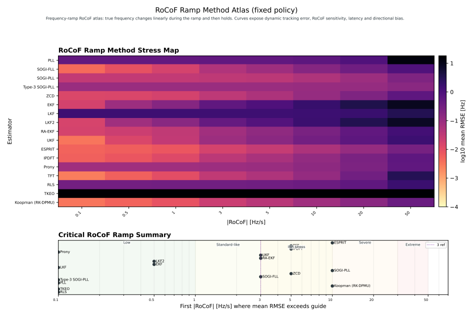
Speaker notes

- This is a diagnostic dense sweep, not a qualification result.
- Worst-case phase jump is not necessarily 180 degrees.
- Many methods peak around 90–120 degrees and recover toward 180 degrees.
- Therefore, qualification should sweep jump angle, not test one nominal point.

No estimator owns the whole stress space

ATLAS VALIDATED SWEEP: $n = 30$, **fixed policy, both signs — readiness gate passed (9/9 sweeps)**

RoCoF regime	Best	med. RMSE
Low (0.1–1 Hz/s)	Koopman	5.5 mHz
Standard (1–3)	ESPRIT	12.2 mHz
IBR stress (3–10)	ESPRIT	33.0 mHz
Severe (10–30)	SOGI-PLL	70.3 mHz



Across stress types

EKF wins phase jumps, the magnitude step and interharmonics; ESPRIT/Koopman win the RoCoF and frequency-step regimes; ZCD wins low-THD harmonics; SOGI-PLL leads severe RoCoF. No method leads everywhere.

The deliverable is a selection map: event class × metric × compute budget.

Source: atlas-papergrade-missing-v1/global_metrics_report.csv; RoCoF regimes averaged acros[illegible] signs

Expanded & ATLAS

No estimator owns stress space

Purpose. Show rotation of champions.

Speaker notes

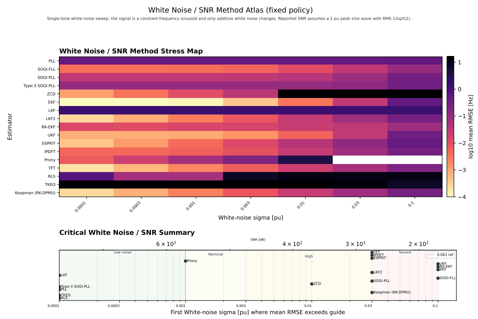
- RoCoF winners rotate by severity regime.
- Across stress types, different estimators win different regions.
- Key line: event class times metric times compute budget is the selection object.

ZCD is a narrow instrument, not a general estimator

ATLAS VALIDATED SWEEP: $n = 30$, **fixed policy, both signs — readiness gate passed (9/9 sweeps)**

Low-stress champion

Up to ∼15% THD, zero-crossing detection gives the *lowest* error of any method — **0.27 mHz** at 1% THD, 2.2 mHz at 15%; and just 1.1 mHz at low broadband noise.



ZCD: low-stress champion

Purpose. Start the paradox slowly.

Speaker notes

- ZCD is excellent in clean high-SNR crossings.
- Up to moderate THD and low broadband noise, it can produce the lowest error.
- Do not dismiss it too early; the next clicks explain the boundary.

ZCD is a narrow instrument, not a general estimator

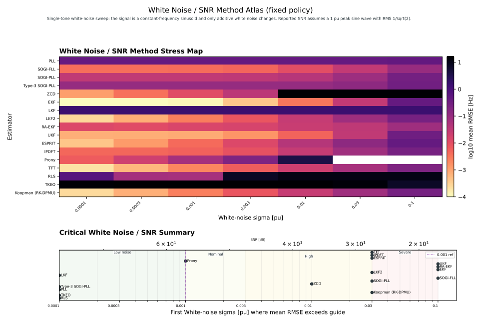
ATLAS VALIDATED SWEEP: $n = 30$**, fixed policy, both signs — readiness gate passed (9/9 sweeps)**

Low-stress champion

Up to ∼15% THD, zero-crossing detection gives the *lowest* error of any method — **0.27 mHz** at 1% THD, 2.2 mHz at 15%; and just 1.1 mHz at low broadband noise.

Broadband-noise cliff

Same method: 34 mHz at $\sigma = 0.003$ pu $\rightarrow$ **3112 Hz** at $\sigma = 0.1$ pu. EKF stays ≤ 0.43 Hz across the same sweep.



ZCD: broadband-noise cliff

Purpose. Reveal the failure mode.

Speaker notes

- The same method collapses under stronger broadband noise.
- EKF remains within a bounded error range while ZCD can explode.
- This is the key lesson: a low-stress winner may be a high-stress failure.

ZCD is a narrow instrument, not a general estimator

ATLAS VALIDATED SWEEP: $n = 30$, **fixed policy, both signs — readiness gate passed (9/9 sweeps)**

Low-stress champion

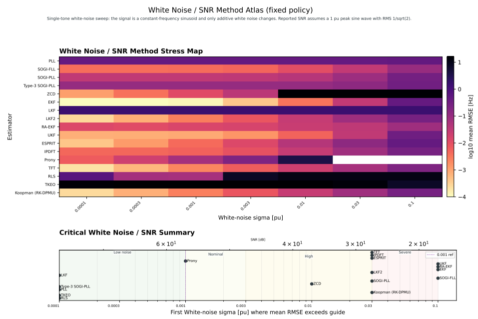
Up to $\sim$15% THD, zero-crossing detection gives the *lowest* error of any method — **0.27 mHz** at 1% THD, 2.2 mHz at 15%; and just 1.1 mHz at low broadband noise.

Broadband-noise cliff

Same method: 34 mHz at $\sigma = 0.003$ pu $\rightarrow$ **3112 Hz** at $\sigma = 0.1$ pu. EKF stays ≤ 0.43 Hz across the same sweep.

Harmonic cliff

Same method again: 2.2 mHz at 15% THD $\rightarrow$ **39 Hz** at 20% $\rightarrow$ **86 Hz** at 30%.



ZCD: harmonic cliff

Purpose. Complete the paradox.

Speaker notes

- ZCD also collapses above about 15 percent THD in this sweep.
- Aggregate rankings hide this kind of stress-specific collapse.
- Keep the conclusion short: ZCD is narrow, not general.

Voltage sags and swells are not symmetric

ATLAS VALIDATED SWEEP: $n = 30$, **fixed policy, both signs — readiness gate passed (9/9 sweeps)**

Estimator at $\pm 90\%$	Sag to 0.1 pu	Swell to 1.9 pu
EKF RMSE	47 mHz	14 mHz
RA-EKF RMSE	49 mHz	30 mHz
ZCD RMSE	374 Hz	9.4 mHz
ZCD peak error	2856 Hz	32 mHz

Mechanism

Zero-cross timing error scales roughly as $\sigma_n/(A\omega)$. A deep sag lowers zero-cross slope and makes false crossings likely when absolute noise is fixed.

Source: atlas-papergrade-missing-v1/global_metrics_report.csv; src/estimators/zcd.py

What to claim

ZCD is amplitude-invariant only in the noise-free ideal case. Under fixed absolute noise, deep voltage sags create a low-SNR crossing problem.

Benchmark consequence

Sag and swell signs must be swept separately; a positive-only magnitude-step test can hide the worst case.

Sag and swell asymmetry

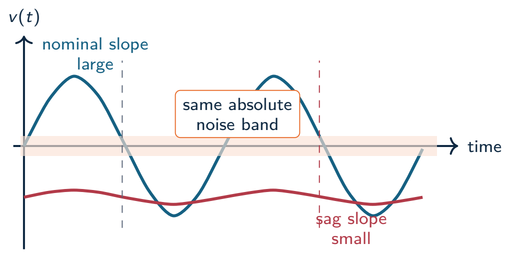
Purpose. Give the theory intuition.

Speaker notes

- ZCD is amplitude-invariant only in the noise-free ideal case.
- With fixed absolute noise, a deep sag lowers zero-crossing slope and creates a low-SNR crossing problem.
- Sag and swell signs must be swept separately.

Why a voltage sag breaks zero-crossing detection

ATLAS VALIDATED SWEEP: $n = 30$, **fixed policy, both signs — readiness gate passed (9/9 sweeps)**



Timing sensitivity

$$\Delta t \approx \frac{n(t_0)}{\left.\frac{dv}{dt}\right|_{t_0}} \qquad \left.\frac{dv}{dt}\right|_0 \approx A\omega$$

When the voltage amplitude A collapses, the same noise produces much larger crossing-time error.

Estimator consequence

ZCD is elegant at clean high-SNR crossings, but deep sags convert amplitude loss into apparent frequency jitter and false crossings.

Why sag breaks ZCD

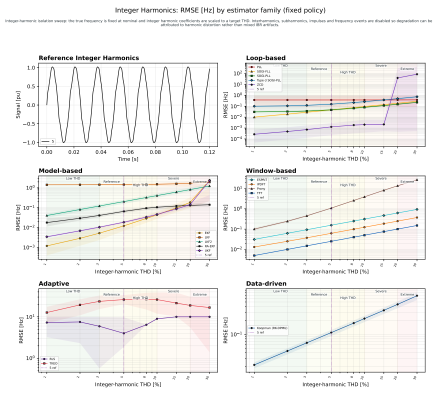
Purpose. Explain the equation physically.

Speaker notes

- Timing error scales as noise divided by the voltage slope at the crossing.
- The slope is proportional to amplitude times angular frequency.
- When amplitude collapses, the same noise creates much larger timing error.

Harmonic degradation differs by estimator family

ATLAS VALIDATED SWEEP: $n = 30$, **fixed policy, both signs — readiness gate passed (9/9 sweeps)**



Result

- Loop, window, model, and adaptive families degrade with THD at different rates.
- Best low-THD accuracy does not predict high-THD survival.
- Loop methods (ZCD) win at low THD but collapse above ∼15%; window methods (TFT) degrade gently and lead at extreme THD.

Harmonic degradation

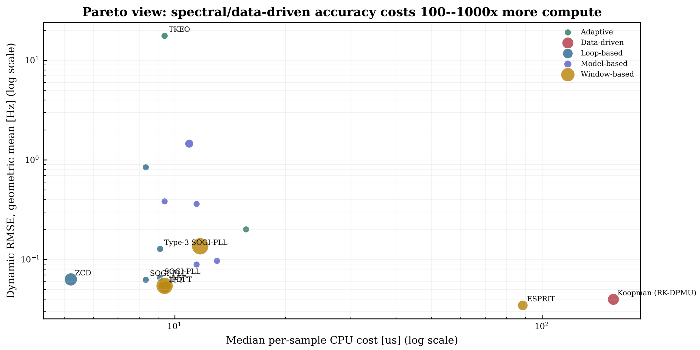
Purpose. Use family slopes, not single points.

Speaker notes

- Loop, window, model, and adaptive families degrade at different THD rates.
- Best low-THD accuracy does not predict high-THD survival.
- Window methods degrade gently and can lead at extreme THD; loop methods can collapse.

Accuracy costs 100–1000× more compute

OPENFREQBENCH EXPANSION: 18 estimators, 33 scenarios, $n = 5$ seeds per scenario



Source: `OpenFreqBench expansion: artifacts/full_mc_benchmark`, $n = 5$

Low-cost core

PLL, SOGI-FLL, EKF, and RA-EKF sit in the deployable compute range.

Accuracy premium

Spectral and data-driven methods can move right by 100–1000× CPU cost.

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CPU-accuracy Pareto

Purpose. Turn the plot into a deployment message.

Speaker notes

- Low-cost core: PLL, SOGI-FLL, EKF, and RA-EKF.
- Accuracy premium: spectral and data-driven methods can move right by 100–1000 times CPU cost.
- Key line: accuracy must be evaluated with trip exposure and compute budget.

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Findings and deployment divider

Purpose. Prepare the conclusion.

Speaker notes

- This section translates results into deployment and qualification language.
- The main result is not a single estimator. It is a selection and validation workflow.
- Keep the pace steady.

Three findings

1. Metric-conditioned
The SGSMA matrix and ATLAS both show that RMSE, trip exposure, RoCoF, and CPU can select different methods.

2. Stress-bounded
AM remains survivable for many methods; severe FM, RoCoF, interharmonics, noise, and extreme THD break the 0.05 Hz guide.

3. Robustness over point accuracy
The OpenFreqBench ramp seeds expose silent divergence; RA-EKF removes that failure mode in the tested run.

> Dynamic frequency estimators should be qualified by event family, sign, severity, recovery, trip exposure, and compute envelope.

Three findings

Purpose. Deliver the thesis in three points.

Speaker notes

- First: estimator choice is metric-conditioned.
- Second: stress boundaries matter; severe stress regimes break the 0.05 Hz guide.
- Third: robustness beats point accuracy; seed-dependent divergence matters.
- Close with the qualification sentence.

Validity limits and next steps

Validation limits

- Limited to the reported scenario set.
- CPU is a software-level cost indicator; hardware latency requires platform timing.
- The study is simulation-only.
- Scenario-wise tuning can affect rank stability.

Validation program

- Hardware-in-the-loop validation.
- Wider converter-control interaction coverage.
- Sensitivity analysis of the tuning protocol.
- ATLAS severity and sign sweeps with validation gates.

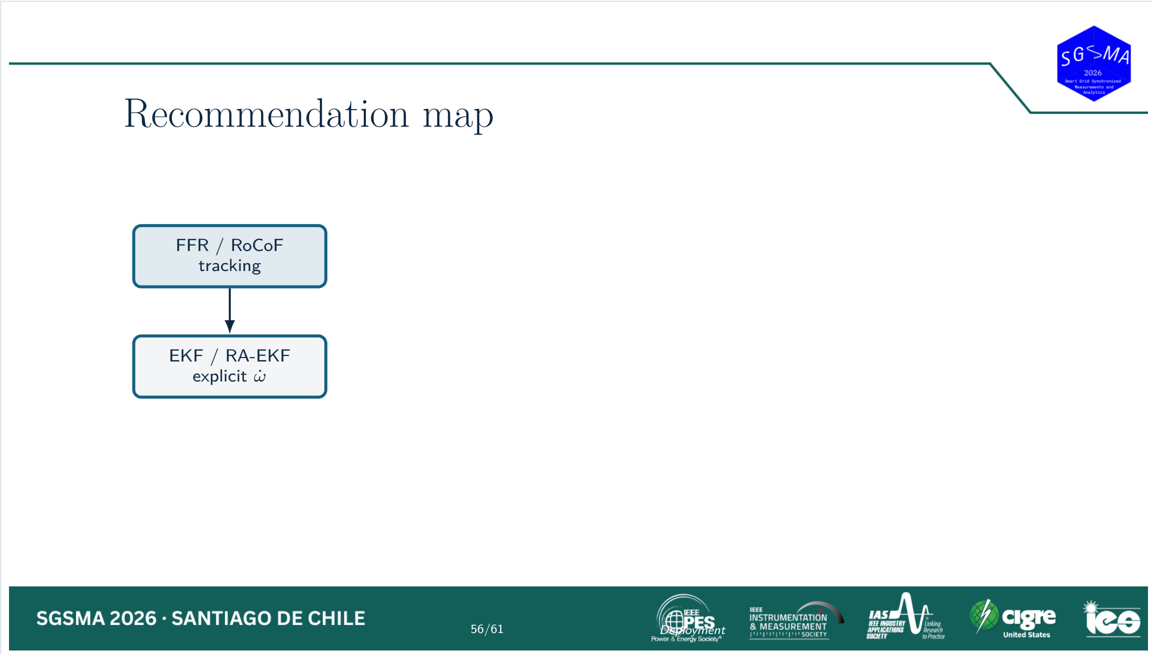
> These limits define the validation path: broader event coverage, hardware timing, fixed-policy replication, and field replay.

Validity limits

Purpose. Be candid and precise.

Speaker notes

- Limits: scenario set, software-level CPU timing, simulation-only evidence, and scenario-wise tuning.
- These are not weaknesses to hide; they define the validation path.
- Transition to recommendation map.



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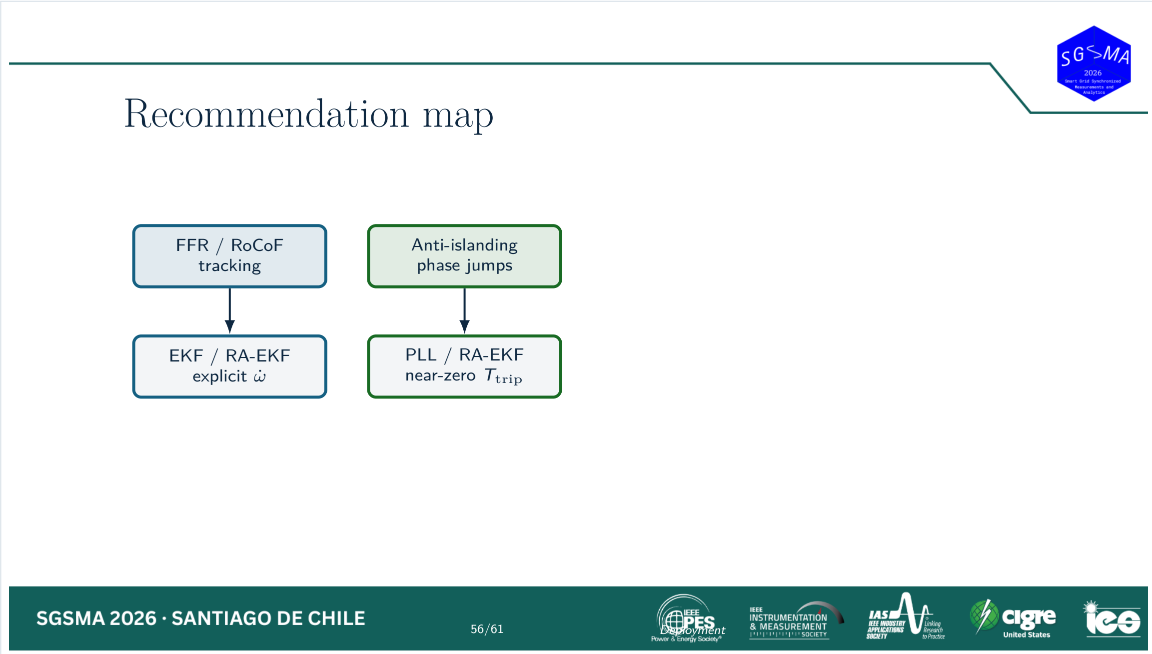
Recommendation map: first layer

Purpose. Start with event class.

Speaker notes

- The map starts with deployment task: FFR/RoCoF, anti-islanding, low-cost relays, or metering-like steady state.
- Say: the event class comes before the estimator choice.
- Let the audience read the top row.

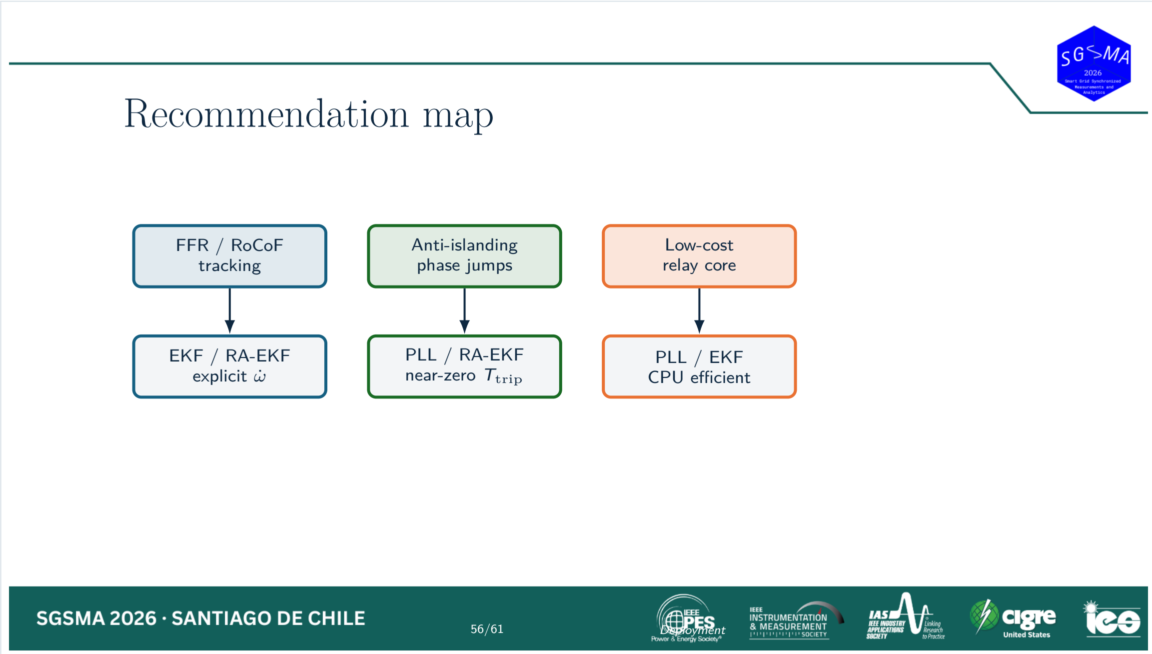


Recommendation map: FFR/RoCoF

Purpose. Explain the first recommendation.

Speaker notes

- For FFR or RoCoF tracking, EKF/RA-EKF are plausible because they model dynamic frequency explicitly.
- This is a conditional recommendation, tied to event class and compute budget.
- Do not say "always deploy RA-EKF."



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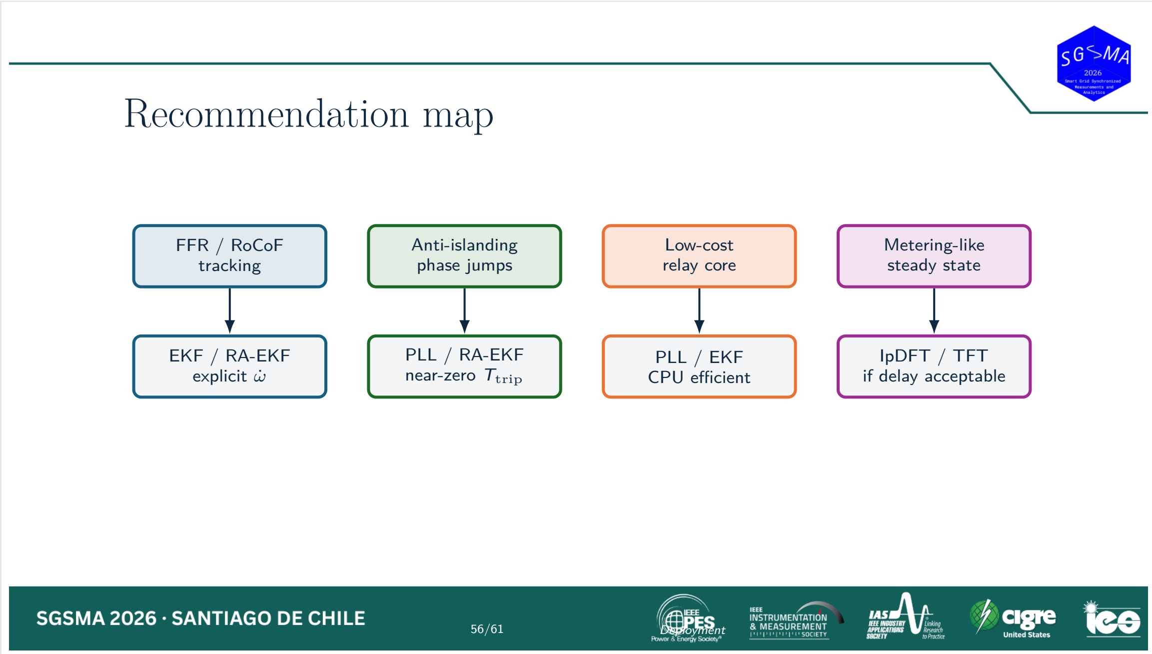
Recommendation map: anti-islanding

Purpose. Explain protection-adjacent use.

Speaker notes

- Anti-islanding and phase jumps require near-zero trip exposure.
- PLL and RA-EKF are highlighted because clean jumps and recovery behavior matter more than one RMSE score.
- Mention that multi-event stress still requires caution.



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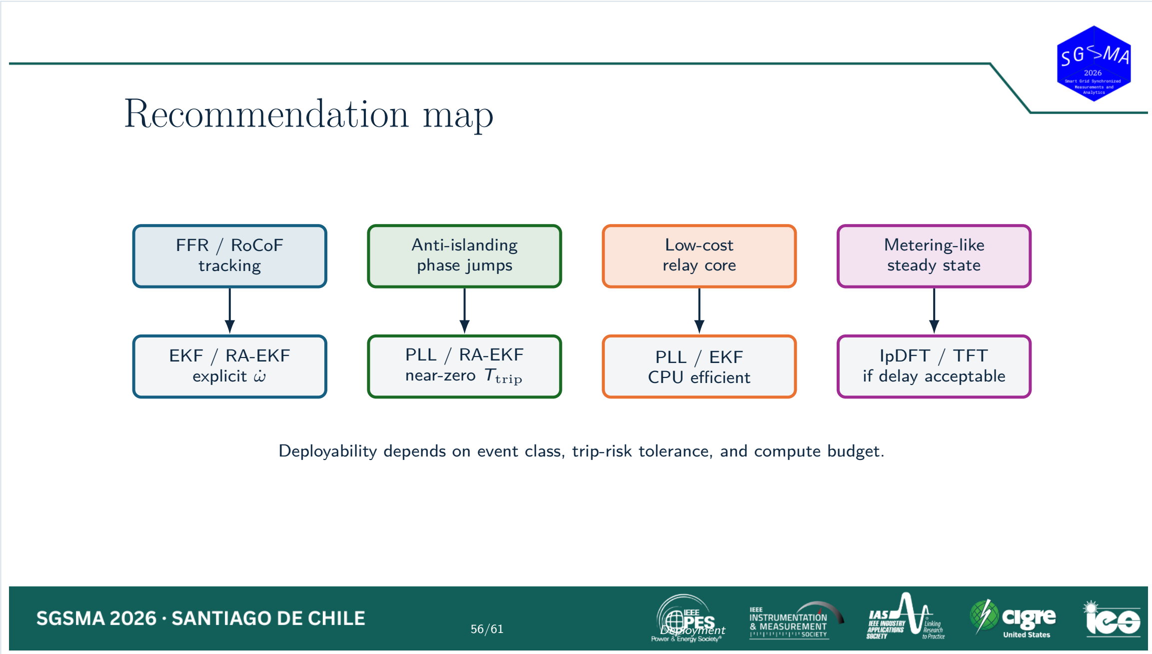
Recommendation map: low-cost relays

Purpose. Explain embedded feasibility.

Speaker notes

- Low-cost relay-core use favors PLL/EKF class methods because compute budget dominates.
- The point is not lowest RMSE; it is acceptable error under strict cost.
- This connects back to CPU feasibility and Pareto slides.

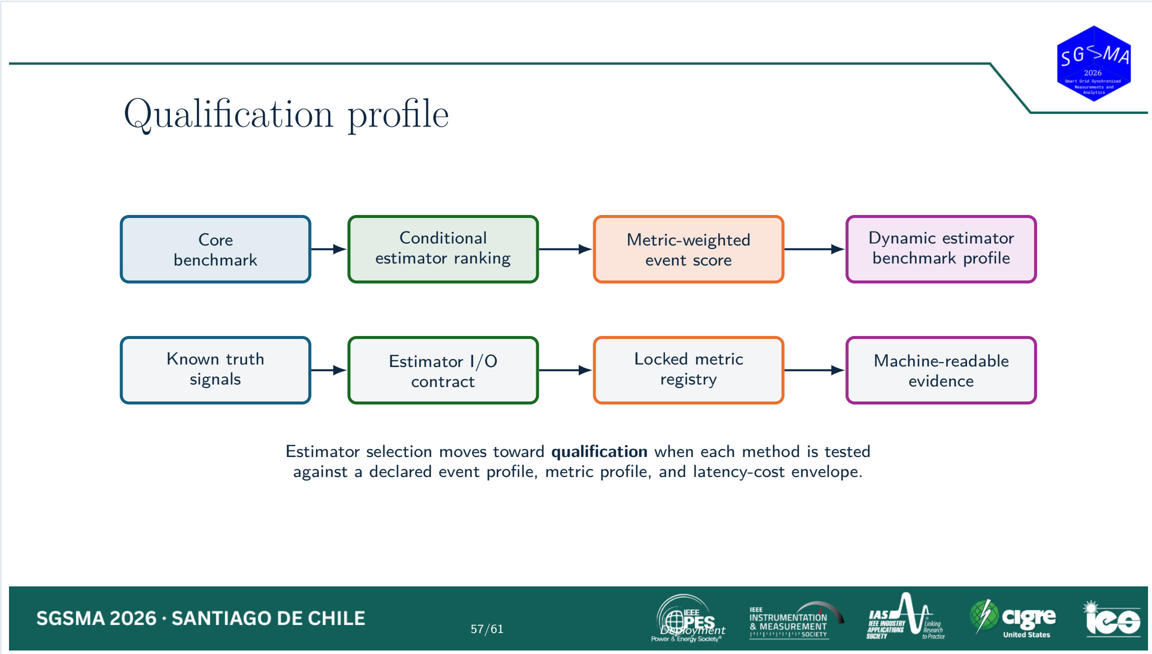


Recommendation map: metering-like state

Purpose. Explain steady-state use.

Speaker notes

- For metering-like steady state, IpDFT/TFT can still be very attractive if delay is acceptable.
- Windowed methods are not bad; they are task-dependent.
- This reinforces the non-universal conclusion.



Qualification profile

Purpose. Define the standardization direction.

Speaker notes

- Selection becomes qualification when each method is tested against a declared event profile, metric profile, and latency-cost envelope.
- The profile needs known truth signals, estimator I/O contract, locked metrics, and machine-readable evidence.
- This is one of the big-picture outputs of the work.

Standardization profile

Current standardization layer

- PMU standards define phasor, frequency, and RoCoF measurement requirements.
- Compliance tests are necessary, with estimator-family ranking by IBR event survival left outside scope.
- Device reports often hide algorithmic failure modes behind aggregate accuracy.

Standardization boundary

A dynamic-estimator profile needs declared signals, estimator I/O, metrics, evidence artifacts, and latency-cost reporting.

Benchmark profile

- Canonical events: steps, ramps, phase jumps, ringdown, harmonics, interharmonics, noise, and composite IBR sequences.
- Locked metrics: RMSE, peak error, RoCoF error, trip-risk, settling, invalid samples, CPU, and latency.
- Task profiles: forensic monitoring, PMU augmentation, DFR support, and relay-adjacent screening.

Standardization value

A reference test profile needs the same signals, estimator I/O, metrics, and machine-readable evidence.

Standardization profile

Purpose. Connect to future standards without overclaiming.

Speaker notes

- Current PMU standards define measurement requirements, but not estimator-family ranking under IBR event survival.
- The benchmark profile adds canonical events, locked metrics, and task profiles.
- Phrase carefully: this can inform a reference test profile; it is not yet a standard.

IBR Event Qualification Score

COMPOSITE EVENT SCORE: weighted event-class qualification metric

Score for estimator e

$$\mathrm{IBR\text{-}ERS}(e) = 1 - \sum_{s \in \mathcal{S}} w_s \, \Phi_s(e),$$

$$\Phi_s(e) = \alpha \tilde{E}_s + \beta \tilde{P}_s + \gamma \tilde{T}_s + \delta \tilde{S}_s + \eta \tilde{C}_s + \lambda \tilde{L}_s.$$

Score composition

RMSE | peak | trip | settle | cost

$\tilde{E}_s, \tilde{P}_s$ dynamic and peak error
$\tilde{T}_s, \tilde{S}_s$ trip exposure and recovery
$\tilde{C}_s, \tilde{L}_s$ compute and latency penalties

Why a score is needed

RMSE, trip-risk, transient recovery, and cost lead to different decisions. A composite score makes the deployment objective explicit.

Protection profile

Raise γ, δ, λ when trip exposure, recovery, and latency dominate the risk model.

Monitoring profile

Raise α, β, η when reconstruction accuracy and compute budget dominate.

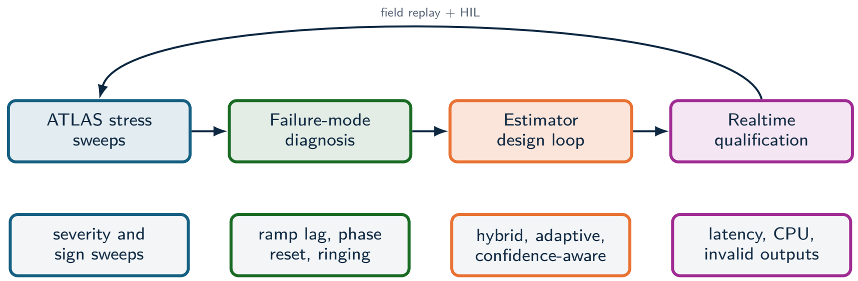
IBR Event Qualification Score

Purpose. Explain the composite metric.

Speaker notes

- The score is a way to state the deployment objective explicitly.
- The same estimator can rank differently if the profile raises trip exposure, recovery, compute, or reconstruction accuracy.
- Keep this as a proposed qualification metric, not a validated standard score.

Toward realtime IBR estimators



- The stress analysis identifies which physical stressor breaks each estimator family.
- That diagnosis supports hybrid estimators: event classifiers, harmonic-aware preprocessing, adaptive bandwidth, and self-reported confidence.
- Validation should combine synthetic truth, EMT replay, hardware-in-the-loop, and field oscillography.

Toward realtime IBR estimators

Purpose. Describe what the results enable.

Speaker notes

- ATLAS identifies which stressor breaks each estimator family.
- That diagnosis supports hybrid estimators: event classifiers, harmonic-aware preprocessing, adaptive bandwidth, and confidence-aware outputs.
- Validation should combine synthetic truth, EMT replay, HIL, and field oscillography.

Validation agenda and conclusion

Conclusion

- Field disturbances define the measurement problem; controlled synthetic truth makes estimator behavior explainable.
- The study compares estimators across objectives, regimes, and deployment constraints, including IBR-relevant stress modes.
- Empirical conclusions remain tied to evidence level, stress regime, and replication depth.

Validation agenda

- Full ATLAS sweeps: fixed policy, 100-run Monte Carlo, statistical rank-stability analysis.
- Composite qualification profiles for forensic, PMU-augmentation, and relay-adjacent tasks.
- Three-phase, unbalance, and converter-control interaction coverage.
- Real-event replay from DFR/PMU/oscillography with synchronized ground-truth proxies.
- Estimators that optimize the full objective vector: frequency error, RoCoF, trip exposure, settling, CPU, latency.

Result: estimator choice is tied to event class, metric target, and compute envelope; estimator qualification must declare the stress regime, latency limit, and evidence depth.

Conclusion

Purpose. End with the single sentence they should remember.

Speaker notes

- Field disturbances define the measurement problem; controlled truth makes estimator behavior explainable.
- The benchmark compares estimators across objectives, regimes, and deployment constraints.
- Closing line: estimator choice is tied to event class, metric target, and compute envelope; qualification must declare stress regime, latency limit, and evidence depth.